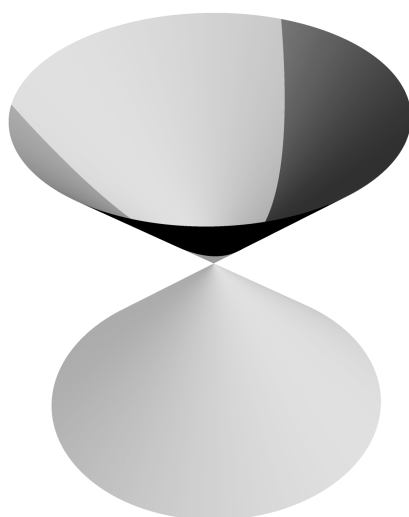


Geometry I

Draft 2026



1	Affine space	3
1.1	Affine space	3
1.2	Cartesian coordinates	7
1.3	Cartesian coordinates as projections	9
1.4	Changing Cartesian frames	10
1.5	Orientation	13
2	Affine subspaces	16
2.1	Lines in $\mathbb{A}^2$	17
2.2	Planes in $\mathbb{A}^3$	20
2.3	Lines in $\mathbb{A}^3$	23
2.4	Affine subspaces of $\mathbb{A}^n$	26
	Appendices	32
A	Changing the basis in a vector space	33
B	Coordinate systems	35
B.1	Polar coordinates	35
B.2	Cylindrical coordinates	36
B.3	Spherical coordinates	36
B.4	Barycentric coordinates	37
C	Pencils of lines and planes	38
C.1	Pencil of lines in $\mathbb{A}^2$	38
C.2	Pencil of planes in $\mathbb{A}^3$	39
D	Some classical theorems in affine geometry	41

Bibliography

46

Contents

1.1 Affine space	3
1.1.1 The vector space structure of geometric vectors	3
1.1.2 Vectors acting on points	5
1.2 Cartesian coordinates	7
1.2.1 Frames in dimension 2	7
1.2.2 Frames in dimension 3 and n	8
1.3 Cartesian coordinates as projections	9
1.4 Changing Cartesian frames	10
1.5 Orientation	13

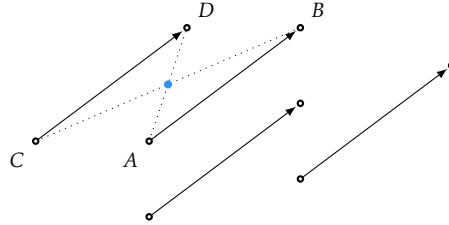
1.1 Affine space

1.1.1 The vector space structure of geometric vectors

We denote by $\mathbb{E}$ the Euclidean space. It describes the Platonic realm of perfect shapes which resonate with our intuition. To explore this world, the ancient Greeks used primitives and axioms in the form of Euclid's postulates [10]. Eventually, it was noticed that a more rigorous description is needed. A revision of the foundations of Euclidean geometry was proposed by Hilbert in his *Foundations of Geometry* [14]. An exploration of the axiomatics of $\mathbb{E}$ can be found in [16]. For now, we assume that $\mathbb{E}$ is a set of *points* governed by the geometric intuition which you have acquired so far in school.

A fundamental notion underlying our current way of understanding Euclidean geometry is the notion of *vector*. To define vectors we use the *equipollence* relation which we introduce next. Notice that a straight line segment $[AB]$, between the point A and the point B , has an *orientation* implicit in

the notation: we mention first A and then B . To take this orientation into account we call $[AB]$ the *oriented line segment from A to B* . Now, two oriented segments $[AB]$ and $[CD]$ are called *equipollent*, and we write $[AB] \sim [CD]$, if the segments $[AD]$ and $[BC]$ have the same midpoints.



For noncollinear points, this is equivalent to $ABDC$ being a parallelogram. Moreover, the equipollence relation is an equivalence relation. The equivalence classes of the equipollence relation are called *vectors*. The vector containing the oriented segment $[AB]$ is traditionally denoted by $\overrightarrow{AB}$. We also use the notation $B - A$ for this vector. Formally, we have

$$B - A = \overrightarrow{AB} := \{ \text{ordered segments } [XY] \text{ such that } [XY] \sim [AB] \}.$$

We say that $[AB]$ is a *representative* of the vector $\overrightarrow{AB}$. If $\overrightarrow{AB} = \overrightarrow{CD}$ we say that $[AB]$ and $[CD]$ *define the same vector*. The set of all vectors defined with points in $\mathbb{E}$ is the set of equivalence classes:

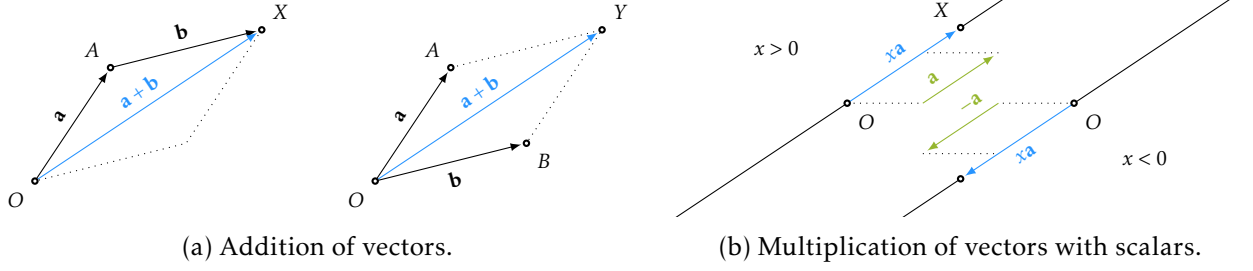
$$\mathbb{V} = \{ \overrightarrow{AB} : \text{where } A \text{ and } B \text{ are points in } \mathbb{E} \} = \mathbb{E} \times \mathbb{E} / \sim.$$

The vector $\overrightarrow{AA}$ is called the *zero vector*, and we denote it by $\vec{0}$ or simply by 0 when there is no risk of confusion. Since all representatives of a vector $\overrightarrow{AB}$ have the same length $|AB|$ this will also be the *length* of the vector $\overrightarrow{AB}$, and we denote it by $|\overrightarrow{AB}|$, equivalently $|B - A|$. This defines a length map $|\square| : \mathbb{V} \rightarrow \mathbb{R}_{\geq 0}$. The vector $\overrightarrow{BA}$ is called the *opposite of the vector $\overrightarrow{AB}$* and is denoted by $-\overrightarrow{AB}$. This defines a map $-\square : \mathbb{V} \rightarrow \mathbb{V}$. Notice that $-(B - A) = A - B$ and that $|\vec{0}| = |\overrightarrow{AA}| = |A - A| = 0$.

Theorem 1.1. The set of vectors $\mathbb{V}$ with vector addition and scalar multiplication is a vector space.

This theorem can be derived from any complete set of axioms describing Euclidean geometry. Here we simply recall the construction of the two operations.

Addition of vectors: Consider two vectors $\mathbf{a}$ and $\mathbf{b}$. If we fix a point O , there is a unique point A such that $\mathbf{a} = \overrightarrow{OA}$ and for the point A there exists a unique point X such that $\mathbf{b} = \overrightarrow{AX}$. The *sum of $\mathbf{a}$ and $\mathbf{b}$* is by definition the vector $\overrightarrow{OX}$, and we denote the sum by $\mathbf{a} + \mathbf{b}$. This defines a map $\square + \square : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{V}$. Equivalently, for a fixed point O there are unique points A and B such that $\mathbf{a} = \overrightarrow{OA}$ and $\mathbf{b} = \overrightarrow{OB}$ and for the points O, A and B there is a *unique* point Y such that $OAYB$ is a parallelogram. It follows that $X = Y$ and therefore $\mathbf{a} + \mathbf{b} = \overrightarrow{OY} = \overrightarrow{OX}$.



Multiplication with scalars: Consider a non-zero vector $\mathbf{a} = \overrightarrow{OA}$ and a scalar $x \in \mathbb{R}$. If $x > 0$, there is a unique point X on the ray from O to A such that $|OX| = x \cdot |OA|$. The *multiplication of the vector $\mathbf{a}$ with the scalar x* , denoted $x \cdot \mathbf{a}$ (or simply $x\mathbf{a}$), defines a map $\square \cdot \square : \mathbb{R} \times \mathbb{V} \rightarrow \mathbb{V}$, and is given by

$$x \cdot \mathbf{a} = \begin{cases} \overrightarrow{OX} & \text{for } \mathbf{a} \neq \overrightarrow{0}, x > 0 \text{ and } X \text{ as above,} \\ -(|x|\mathbf{a}) & \text{for } \mathbf{a} \neq \overrightarrow{0}, x < 0, \\ \overrightarrow{0} & \text{for } \mathbf{a} = \overrightarrow{0} \text{ or } x = 0. \end{cases}$$

1.1.2 Vectors acting on points

Above we have described geometric vectors and their vector space structure. Points in $\mathbb{E}$ and vectors in $\mathbb{V}$ interact in two ways. Firstly, for any fixed point O , there is a bijection $\mathbb{E} \leftrightarrow \mathbb{V}$ which allows us to identify points with vectors. Indeed, for any point A there is a unique vector from O to A . The vector $\overrightarrow{OA}$ is called the *position vector of A relative to O* or simply the *position vector of A* if it is clear from the context what O is. It is customary to denote this vector by $\mathbf{r}_O(A)$ (the $\mathbf{r}$ stands for *radius vector*). Secondly, we may use vectors to ‘move’ points. Namely, for a vector $\mathbf{a}$ and a point A , there is a unique point B such that $\mathbf{a} = \overrightarrow{AB}$. We may think about this as saying that the vector $\mathbf{a}$ is moving the point A to the point B . In technical terms we say that *vectors act on points*. This interaction between points and vectors can be encapsulated in a data structure called *affine space*.

Definition 1.2. An affine space $\mathbb{A}$ is a set of points to which we attach a vector space $\mathbb{V}$ and two maps

$$\square - \square : \mathbb{A} \times \mathbb{A} \rightarrow \mathbb{V} \quad \text{and} \quad \square + \square : \mathbb{V} \times \mathbb{A} \rightarrow \mathbb{A}$$

subject to the following axioms:

- (AS1) For any point P we have $\overrightarrow{0} + P = P$
- (AS2) For any point P and any vectors $\mathbf{v}, \mathbf{w}$ we have $(\mathbf{v} + \mathbf{w}) + P = \mathbf{v} + (\mathbf{w} + P)$
- (AS3) For any two points A and B we have $(A - B) + B = A$
- (AS4) For any point P and any vector $\mathbf{v}$ we have $(\mathbf{v} + P) - P = \mathbf{v}$

The first map produces a vector from two points. It overlaps with the notation $A - B$ for the vector $\overrightarrow{AB}$ mentioned in the previous section. The second map is called *translation* since it takes a vector $\mathbf{v}$ and a point A , and it produces the point B such that $\mathbf{v} = \overrightarrow{AB}$.

The vector space $\mathbb{V}$ attached to the affine space $\mathbb{A}$ is denoted by $D(\mathbb{A})$ and is called the *direction space* of $\mathbb{A}$. The dimension of $\mathbb{A}$ is by definition the dimension of $D(\mathbb{A})$ and is denoted by $\dim \mathbb{A}$. It is customary to fix the dimension and indicate it with an upper index, e.g., we write $\mathbb{A}^n$ for an n -dimensional affine space.

Example 1.3. The term *affine* was coined by Leonhard Euler to refer to certain arguments and properties of certain geometric shapes which do not change when stretching or contracting the Euclidean space. We will make this more precise in a later chapter. An affine space is a first data structure that captures a considerable part of Euclidean geometry. One can deduce from Hilbert's axioms that the Euclidean space $\mathbb{E}$ with set of vectors $D(\mathbb{E}) = \mathbb{V}$, defined in the previous section, is an affine space of dimension at least 3. However, when we want to talk about a particular dimension, we write it as an upper index, e.g., we write $\mathbb{E}^2$ for the Euclidean plane, $\mathbb{E}^3$ for the Euclidean space. The reframing of the geometric canvas as an affine space unlocks the full power of linear algebra. From Hilbert's axioms one can deduce in particular the following facts.

Definition 1.4. Let $\mathbb{A}$ be an affine space. A subset $S \subseteq \mathbb{A}$ is called an *affine subspace* if there exists a point $O \in S$ and a vector subspace $\mathbb{U} \subseteq \mathbb{V}$ such that

$$S = \mathbb{U} + O = \{\mathbf{u} + O : \mathbf{u} \in \mathbb{U}\}.$$

In this case, the subspace $\mathbb{U}$ is unique and is denoted by $D(S)$. It is called the *direction subspace* of S . The dimension of S is defined as $\dim(S) = \dim(D(S))$.

- A *point* is an affine subspace of dimension 0.
- A *line* is an affine subspace of dimension 1.
- A *plane* is an affine subspace of dimension 2.
- A *hyperplane* is an affine subspace of dimension $\dim(\mathbb{A}) - 1$.

For the Euclidean space $\mathbb{E}$, these properties can be deduced from Hilbert's axioms.

Example 1.5. One can deduce from the axioms of an affine space that for any fixed point O there is a bijection between vectors and points given by $\mathbf{v} \mapsto \mathbf{v} + O$. This allows one to identify, element-wise, vectors with points. Vector spaces and affine spaces are in fact in bijection since for any (real) vector space $\mathbb{V}$ one can construct an affine space $\mathbb{A}$ such that $D(\mathbb{A}) = \mathbb{V}$. Indeed, the maps giving the affine space structure are subtraction and addition in the vector space. In this set-up we view the elements of $\mathbb{V}$ in two distinct ways: as points and as vectors which act on points. A vector space has an origin, the zero vector, but on a line or in a plane all points are equal. Marcel Berger [4, Chapter 2] summarizes this relationship nicely by stating that “an affine space is nothing more than a vector space whose origin we try to forget about, by adding translations to the linear maps.”

Remark. The mathematical library *Mathlib* [23] written in Lean [22] defines affine spaces as above in order to treat Euclidean geometry (see [24]).

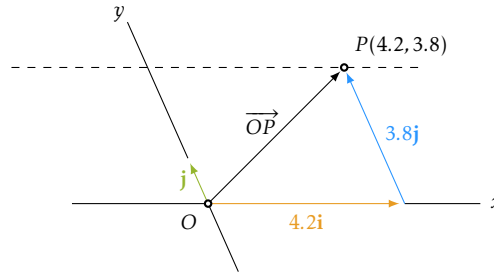
1.2 Cartesian coordinates

Coordinates are tuples of numbers associated with points in a given object or space. In this chapter, we focus on Cartesian frames, named after René Descartes. While modern views on the extent of Descartes' revolution vary, the utility of coordinate geometry in bridging geometry and algebra is undeniable [15, Section 6.8]. For other types of coordinate systems, see Appendix B.

1.2.1 Frames in dimension 2

In the Euclidean plane $\mathbb{E}^2$, fix two lines ℓ_1 and ℓ_2 which intersect at a unique point O . We can describe any point P in $\mathbb{E}^2$ as follows: By Definition 1.4, $D(\ell_1)$ and $D(\ell_2)$ are 1-dimensional, linearly independent vector subspaces of the 2-dimensional vector space $\mathbb{V}^2 = D(\mathbb{E}^2)$. If we select non-zero vectors $\mathbf{i} \in D(\ell_1)$ and $\mathbf{j} \in D(\ell_2)$, then $(\mathbf{i}, \mathbf{j})$ is a basis of $\mathbb{V}^2$. Consequently, there exist *unique* scalars $x_P, y_P \in \mathbb{R}$ such that

$$\mathbf{r}_O(P) = \overrightarrow{OP} = x_P \mathbf{i} + y_P \mathbf{j}. \quad (1.1)$$



This establishes a bijection $\mathbb{E}^2 \leftrightarrow \mathbb{R}^2$ between points in the plane and ordered pairs of real numbers. This bijection arises as the composition of two bijections: 1. the bijection $\mathbf{r}_O : \mathbb{E}^2 \rightarrow \mathbb{V}^2$, which assigns to each point P its position vector $\overrightarrow{OP}$, and 2. the decomposition of vectors $\mathbf{r}_O(P)$ with respect to a basis, establishing the identification $\mathbb{V}^2 \cong \mathbb{R}^2$. Thus, the bijection $\mathbb{E}^2 \leftrightarrow \mathbb{R}^2$ is the composition of a bijection $\mathbb{E}^2 \leftrightarrow \mathbb{V}^2$ and a bijection $\mathbb{V}^2 \leftrightarrow \mathbb{R}^2$, which depend on two choices: 1. selecting a point O for the map $\mathbf{r}_O$, and 2. choosing two vectors $\mathbf{i}$ and $\mathbf{j}$ that form a basis of $\mathbb{V}^2$.

Definition 1.6. A *frame* in $\mathbb{E}^2$ is a pair $\mathcal{K} = (O, \mathcal{B})$, where O is a point in $\mathbb{E}^2$ and $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ is a basis of $\mathbb{V}^2$. A frame is also referred to as a *Cartesian coordinate system*, or a *Cartesian frame*. We use the shorter term for convenience. Given a frame $\mathcal{K}$, the unique pair (x_P, y_P) in (1.1) is called the *coordinates of P with respect to the frame $\mathcal{K}$* . In other words, the coordinates of P are the components of the position vector $\mathbf{r}_O(P) = \overrightarrow{OP}$ of P relative to O in the basis $\mathcal{B}$. We write $P_{\mathcal{K}}(x_P, y_P)$ when we want to indicate the coordinates. If it is clear from the context what $\mathcal{K}$ is, we omit the subscript $\mathcal{K}$ and simply write $P(x_P, y_P)$. By convention, the first coordinate is typically denoted x , and the second y . The line ℓ_1 is called the x -axis, denoted Ox , while the line ℓ_2 is the y -axis, denoted Oy . The point O is referred to as the *origin of the frame $\mathcal{K}$* .

For computational purposes, points $P_{\mathcal{K}}(x_P, y_P)$ and vectors $\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j}$ are identified with column matrices, using their coordinates and components, respectively:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_P \\ y_P \end{bmatrix} \in \mathbb{R}^2 \quad \text{and} \quad [\mathbf{a}]_{\mathcal{B}} = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \in \mathbb{R}^2.$$

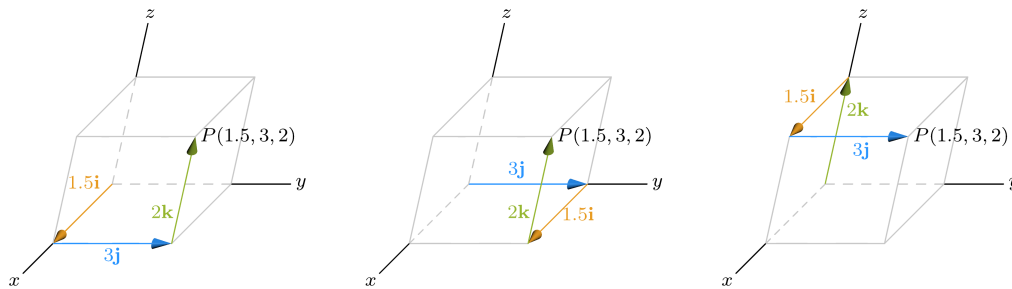
Here, the subscript $\mathcal{K}$ indicates the frame relative to which P has the indicated coordinates, while the subscript $\mathcal{B}$ indicates the basis in which the components of the vector $\mathbf{a}$ are expressed.

1.2.2 Frames in dimension 3 and n

The concepts introduced for dimension 2 generalize naturally to dimension 3 and, more generally, to any dimension n .

In $\mathbb{E}^3$, we fix three non-coplanar lines intersecting at a point O . These lines determine three linearly independent directions in $\mathbb{V}^3 = D(\mathbb{E}^3)$. Choosing basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ along these directions gives us a basis $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$. Any point $P \in \mathbb{E}^3$ is uniquely determined by the decomposition of its position vector:

$$\overrightarrow{OP} = x_P \mathbf{i} + y_P \mathbf{j} + z_P \mathbf{k}. \quad (1.2)$$



Definition 1.7. A *frame* in $\mathbb{A}^n$ (and specifically in $\mathbb{E}^3$ for $n = 3$) is a pair $\mathcal{K} = (O, \mathcal{B})$, where O is a point in the affine space and $\mathcal{B} = (\mathbf{i}_1, \dots, \mathbf{i}_n)$ is a basis of the direction space. The *coordinates* of a point P with respect to $\mathcal{K}$ are the unique scalars $(x_1, \dots, x_n)$ such that

$$\overrightarrow{OP} = x_1 \mathbf{i}_1 + \dots + x_n \mathbf{i}_n.$$

We write $P(x_1, \dots, x_n)$. For $n = 3$, we typically use the notation $P(x, y, z)$ and the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. The lines determined by the basis vectors passing through O are the *coordinate axes* (Ox, Oy, Oz) and the planes determined by pairs of axes are *coordinate planes* (Oxy, Oyz, Ozx).

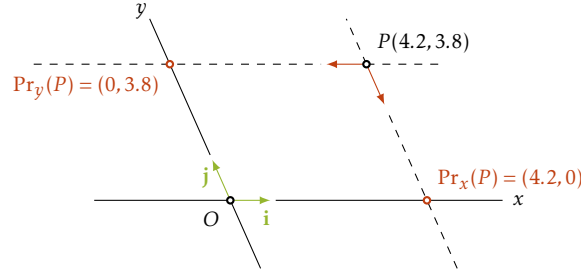
Points and vectors are identified with column matrices of coordinates and components respectively:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad \text{and} \quad [\mathbf{a}]_{\mathcal{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n.$$

1.3 Cartesian coordinates as projections

We explore projections extensively in Chapter ?? . Here we simply observe that the coordinates can be interpreted as projections onto coordinate axes.

In $\mathbb{E}^2$, a point $P(x_P, y_P)$ satisfies $\overrightarrow{OP} = x_P \mathbf{i} + y_P \mathbf{j}$. This is equivalent to the existence of unique points $X \in Ox$ and $Y \in Oy$ such that $OXPY$ is a parallelogram.



Definition 1.8. With the above notation we may define the following maps

$$\text{Pr}_x : \mathbb{E}^2 \rightarrow Ox, \quad \text{Pr}_x(P) = X(x_P, 0) \quad \text{and} \quad \text{Pr}_y : \mathbb{E}^2 \rightarrow Oy, \quad \text{Pr}_y(P) = Y(0, y_P).$$

The map Pr_x is called *the projection on Ox along Oy*, and Pr_y is called *the projection on Oy along Ox*. For vectors $\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j}$, we have similar maps defined as follows

$$\text{pr}_x : \mathbb{V}^2 \rightarrow \mathbb{R}, \quad \text{pr}_x(\mathbf{a}) = a_x \quad \text{and} \quad \text{pr}_y : \mathbb{V}^2 \rightarrow \mathbb{R}, \quad \text{pr}_y(\mathbf{a}) = a_y.$$

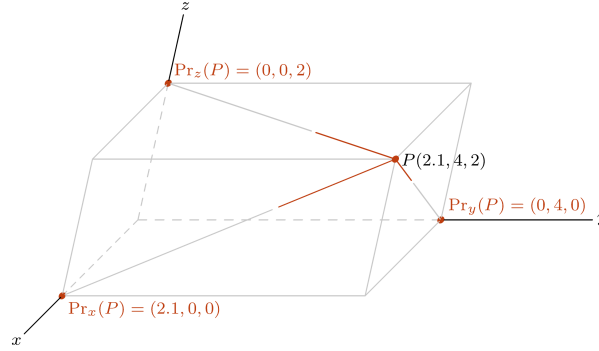
The map pr_x is called *the projection on the first component*, and pr_y is called *the projection on the second component*. From the definitions we immediately deduce the following identities

$$\overrightarrow{O\text{Pr}_x(P)} = \text{pr}_x(\overrightarrow{OP})\mathbf{i}, \quad \overrightarrow{O\text{Pr}_y(P)} = \text{pr}_y(\overrightarrow{OP})\mathbf{j}, \quad \overrightarrow{OP} = \text{pr}_x(\overrightarrow{OP})\mathbf{i} + \text{pr}_y(\overrightarrow{OP})\mathbf{j} \quad \text{and}$$

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_P \\ y_P \end{bmatrix} = \begin{bmatrix} \text{pr}_x(\overrightarrow{OP}) \\ \text{pr}_y(\overrightarrow{OP}) \end{bmatrix} = [\overrightarrow{OP}]_{\mathcal{B}}.$$

Remark. If $\mathbf{i}$ and $\mathbf{j}$ are unit vectors, then for a point P as above, we have $|OX| = x_P, |OY| = y_P$. In other words, the coordinates of P are the lengths of the sides of the parallelogram $OXPY$.

In $\mathbb{E}^3$, for $P(x_P, y_P, z_P)$, there exist unique points $X \in Ox, Y \in Oy, Z \in Oz$ such that O, X, Y, Z, P are vertices of a parallelepiped.



Definition 1.9. For vectors $\mathbf{a} = a_1 \mathbf{i}_1 + \dots + a_n \mathbf{i}_n$ in $D(\mathbb{A}^n)$ we have *projection maps* $\text{pr}_1, \dots, \text{pr}_n : D(\mathbb{A}^n) \rightarrow \mathbb{R}$ defined by $\text{pr}_k(\mathbf{a}) = a_k$. For points P in $\mathbb{A}^n$, we define projections onto the coordinate axes $\text{Pr}_k : \mathbb{A}^n \rightarrow O x_k$ such that $\overrightarrow{O \text{Pr}_k(P)} = \text{pr}_k(\overrightarrow{OP}) \mathbf{i}_k$.

In $\mathbb{E}^2$, Pr_x is the projection on Ox along Oy . In $\mathbb{E}^3$, Pr_x is the projection on Ox along the plane Oyz . From the definitions we immediately deduce:

$$[P]_{\mathcal{K}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \text{pr}_1(\overrightarrow{OP}) \\ \vdots \\ \text{pr}_n(\overrightarrow{OP}) \end{bmatrix} = [\overrightarrow{OP}]_{\mathcal{B}}.$$

1.4 Changing Cartesian frames

A frame $\mathcal{K}$ allows us to identify the set of points in $\mathbb{A}^n$ with $\mathbb{R}^n$. However, an n -tuple of numbers has no geometric meaning in the absence of a frame. Moreover, with respect to different frames, points have different coordinates. The process of translating from one Cartesian frame to another is made precise with the following theorem. Recall from linear algebra the concept of change of basis matrix (see Appendix A).

Theorem 1.10. Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{A}^n$. For any point $P \in \mathbb{A}^n$ we have

$$[P]_{\mathcal{K}'} = M_{\mathcal{B}', \mathcal{B}} \cdot ([P]_{\mathcal{K}} - [O']_{\mathcal{K}}) = M_{\mathcal{B}', \mathcal{B}}^{-1} \cdot ([P]_{\mathcal{K}} - [O']_{\mathcal{K}}) = M_{\mathcal{B}', \mathcal{B}} \cdot [P]_{\mathcal{K}} + [O]_{\mathcal{K}'}. \quad (1.3)$$

Proof. Since the coordinates of a point are the components of its position vector, Equation (1.3) is equivalent to:

$$[\overrightarrow{O'P}]_{\mathcal{B}'} = M_{\mathcal{B}', \mathcal{B}} \cdot ([\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}) = M_{\mathcal{B}', \mathcal{B}}^{-1} \cdot ([\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}) = M_{\mathcal{B}', \mathcal{B}} \cdot [\overrightarrow{OP}]_{\mathcal{B}} + [\overrightarrow{O'O}]_{\mathcal{B}'}. \quad (1.4)$$

Since $[\overrightarrow{O'P}]_{\mathcal{B}} = [\overrightarrow{OP}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}$ and since $M_{\mathcal{B}', \mathcal{B}}$ is the change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$, the first equality follows. The second equality follows from the property that $M_{\mathcal{B}', \mathcal{B}} = M_{\mathcal{B}, \mathcal{B}'}^{-1}$ (see Appendix A). The last equality is obtained by opening the parentheses and noticing that

$$M_{\mathcal{B}, \mathcal{B}'}^{-1} [\overrightarrow{OO'}]_{\mathcal{B}} = [\overrightarrow{OO'}]_{\mathcal{B}'} = -[\overrightarrow{O'O}]_{\mathcal{B}'}.$$

□

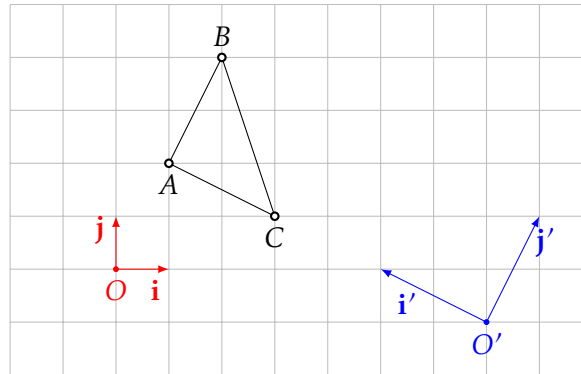
Let us flesh out the steps needed to translate coordinates from one Cartesian frame to another, by expanding Theorem 1.10 using Appendix A. Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{A}^n$ with $\mathcal{B} = (\mathbf{i}_1, \dots, \mathbf{i}_n)$ and $\mathcal{B}' = (\mathbf{i}'_1, \dots, \mathbf{i}'_n)$. Suppose we know $\mathcal{K}'$ in terms of $\mathcal{K}$, i.e., the coordinates of O' relative to $\mathcal{K}$ and the components of $\mathbf{i}'_1, \dots, \mathbf{i}'_n$ with respect to $\mathcal{B}$ are given. If the coordinates of a point P are given relative to $\mathcal{K}$, we can find the coordinates of P relative to $\mathcal{K}'$ with the following steps:

1. Construct the change of basis matrix $M_{\mathcal{B}, \mathcal{B}'}$ by placing $[\mathbf{i}'_1]_{\mathcal{B}}, \dots, [\mathbf{i}'_n]_{\mathcal{B}}$ in the columns of the matrix.
2. Calculate $M_{\mathcal{B}', \mathcal{B}}$ by inverting $M_{\mathcal{B}, \mathcal{B}'}$.
3. Calculate $[P]_{\mathcal{K}'} = M_{\mathcal{B}', \mathcal{B}} \cdot ([P]_{\mathcal{K}} - [O']_{\mathcal{K}})$ since $[P]_{\mathcal{K}}$ and $[O']_{\mathcal{K}}$ are given.

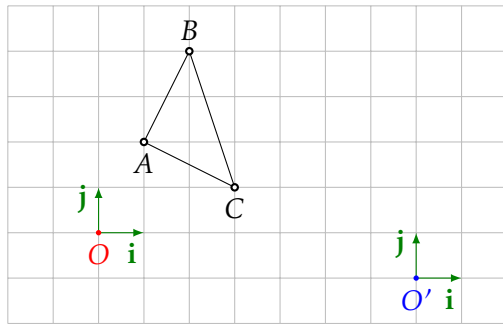
Example 1.11 (In dimension 2). Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{E}^2$, with $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and $\mathcal{B}' = (\mathbf{i}', \mathbf{j}')$. Assume that $O', \mathbf{i}'$ and $\mathbf{j}'$ are known relative to $\mathcal{K}$:

$$[O']_{\mathcal{K}} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \quad (1.5)$$

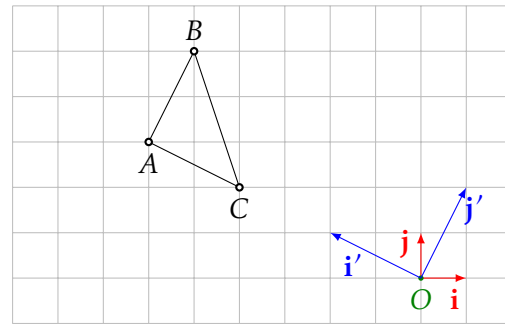
How can we translate the coordinates of points from $\mathcal{K}$ to $\mathcal{K}'$?



We can achieve this in two steps: (a) first, we change the origin, i.e., we go from $(O, \mathcal{B})$ to $(O', \mathcal{B})$, and (b) we change the direction of the coordinate axes, i.e., we go from $(O', \mathcal{B})$ to $(O', \mathcal{B}')$. The first step is simply a translation, while the second corresponds to the usual base change from linear algebra.



(a) Change the origin.



(b) Change the direction of the axes.

In the first step, when going from $\mathcal{K} = (O, \mathcal{B})$ to $\tilde{\mathcal{K}} = (O', \mathcal{B})$, we are looking for the components of the position vector of A relative to O' with respect to $\mathcal{B}$.

$$\overrightarrow{O'A} = \overrightarrow{OA} - \overrightarrow{OO'} \quad \text{and therefore} \quad [\overrightarrow{O'A}]_{\mathcal{B}} = [\overrightarrow{OA}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}.$$

In the second step, when going from $\tilde{\mathcal{K}} = (O', \mathcal{B})$ to $\mathcal{K}' = (O', \mathcal{B}')$, we are looking for the components of the position vector of A relative to O' with respect to $\mathcal{B}'$. From linear algebra, we know that this is done with the change of basis matrix (see Appendix A). Let $M_{\mathcal{B}', \mathcal{B}}$ be the change of basis matrix from the basis $\mathcal{B}$ to the basis $\mathcal{B}'$. Then,

$$[\overrightarrow{OA}]_{\mathcal{B}'} = M_{\mathcal{B}', \mathcal{B}} [\overrightarrow{OA}]_{\mathcal{B}}.$$

Thus, composing the two operations, (a) and (b), we obtain

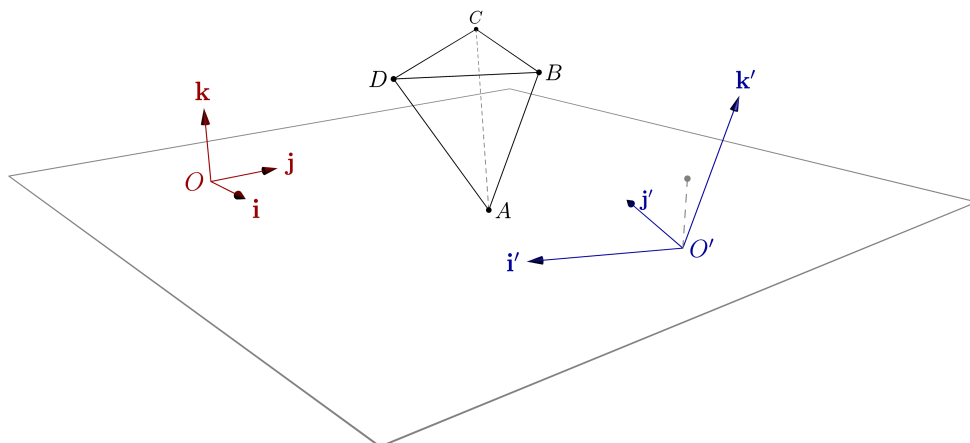
$$[A]_{\mathcal{K}'} = [\overrightarrow{O'A}]_{\mathcal{B}'} = M_{\mathcal{B}', \mathcal{B}} \cdot [\overrightarrow{O'A}]_{\mathcal{B}} = M_{\mathcal{B}', \mathcal{B}} \cdot ([\overrightarrow{OA}]_{\mathcal{B}} - [\overrightarrow{OO'}]_{\mathcal{B}}) \quad (1.6)$$

Now, suppose that A has coordinates $(1, 2)$ with respect to the frame $\mathcal{K}$. Since $[\overrightarrow{OO'}]_{\mathcal{B}} = [O']_{\mathcal{K}}$, by (1.5), we already know the terms in the parentheses on the right-hand side of (1.6). Furthermore, from our assumptions (1.5), it is easy to write down the matrix $M_{\mathcal{B}', \mathcal{B}}$ and then $M_{\mathcal{B}', \mathcal{B}} = M_{\mathcal{B}, \mathcal{B}'}^{-1}$. Thus

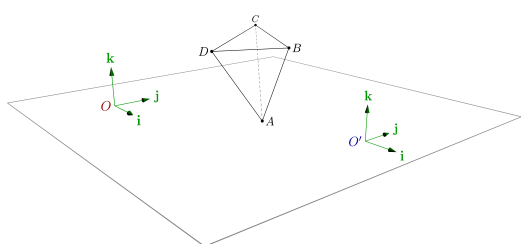
$$[A]_{\mathcal{K}'} = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \cdot \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \frac{1}{-5} \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix} \cdot \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Example 1.12. Let $\mathcal{K} = (O, \mathcal{B})$ and $\mathcal{K}' = (O', \mathcal{B}')$ be two frames in $\mathbb{E}^3$ with $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ and $\mathcal{B}' = (\mathbf{i}', \mathbf{j}', \mathbf{k}')$. Assume that $O', \mathbf{i}', \mathbf{j}'$ and $\mathbf{k}'$ are known relative to $\mathcal{K}$:

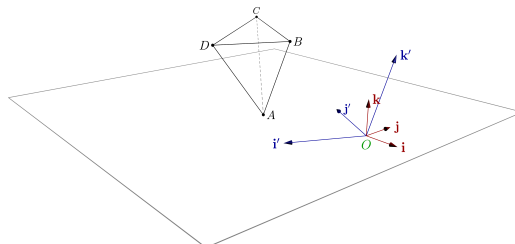
$$[O']_{\mathcal{K}} = \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix}, \quad \mathbf{i}' = -\mathbf{i} - 2\mathbf{j} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{j}' = -2\mathbf{i} + \mathbf{j} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}' = \mathbf{j} + 2\mathbf{k} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$



In any dimension, the coordinates with respect to one frame can be obtained from the coordinates with respect to another frame in two steps.



(a) Change the origin.



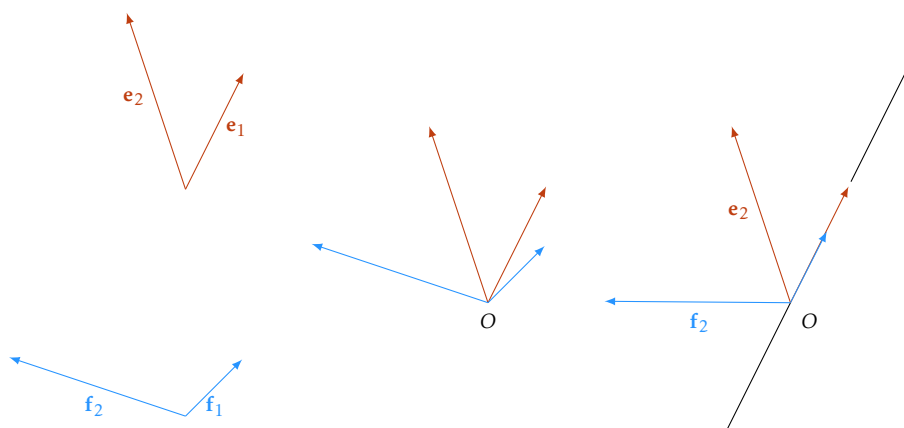
(b) Change the direction of the axes.

Let B be the point with coordinates $(1, 5, 1)$ relative to $\mathcal{K}$. The argument used for dimension 2 (Example 1.5) literally translates to our 3-dimensional setting and we have

$$[B]_{\mathcal{K}'} = M_{B, B'}^{-1} \cdot ([B]_{\mathcal{K}} - [O']_{\mathcal{K}}) = \begin{bmatrix} -1 & -2 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}^{-1} \left(\begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix} \right) = \frac{1}{10} \begin{bmatrix} -2 & -4 & 2 \\ -4 & 2 & -1 \\ 0 & 0 & 5 \end{bmatrix} \left(\begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

1.5 Orientation

Consider the bases $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ and $\mathcal{F} = (\mathbf{f}_1, \mathbf{f}_2)$ of $\mathbb{V}^2$. Represent all four vectors at a common point O and rotate the basis $\mathcal{F}$ such that the vector $\mathbf{f}_1$ points in the same direction as the vector $\mathbf{e}_1$, i.e., such that $\mathbf{f}_1 = \lambda \mathbf{e}_1$ for some positive scalar λ .



The line passing through O with direction vector $\mathbf{e}_1$ (and also $\mathbf{f}_1$) separates the plane $\mathbb{E}^2$ into two half-planes. Considering the positioning of the second vectors, two things can happen: $\mathbf{e}_2$ and $\mathbf{f}_2$ point toward the same half-plane or they point toward different half-planes.

Proposition 1.13. Let $\mathcal{E}$ and $\mathcal{F}$ be as above.

1. If $\det(M_{\mathcal{E},\mathcal{F}}) > 0$ then $\mathbf{e}_2$ and $\mathbf{f}_2$ point to the same half-plane.
2. If $\det(M_{\mathcal{E},\mathcal{F}}) < 0$ then $\mathbf{e}_2$ and $\mathbf{f}_2$ point to different half-planes.

Sketch of the proof. In the above process, we changed the basis $\mathcal{F} = (\mathbf{f}_1, \mathbf{f}_2)$ to the basis $\mathcal{F}' = (\mathbf{f}'_1, \mathbf{f}'_2)$ with a rotation. So, the change of basis matrix $M_{\mathcal{F}',\mathcal{F}}$ is a 2×2 rotation matrix which has determinant equal to 1 (for more on rotations, see Chapter ??). Now, considering the coordinates of the vectors in $\mathcal{F}'$ with respect to $\mathcal{E}$, we have $\mathbf{f}'_1 = (\lambda, 0)$ and $\mathbf{f}'_2 = (a, b)$. Thus, we notice that the vectors $\mathbf{e}_1$ and $\mathbf{f}_1$ point into the same half-plane if $b > 0$. If we now calculate the above determinant, we obtain

$$\det(M_{\mathcal{E},\mathcal{F}}) = \det(M_{\mathcal{E},\mathcal{F}'} \cdot M_{\mathcal{F}',\mathcal{F}}) = \det(M_{\mathcal{E},\mathcal{F}'}) \cdot \underbrace{\det(M_{\mathcal{F}',\mathcal{F}})}_{=1} = \begin{vmatrix} \lambda & a \\ 0 & b \end{vmatrix} = \lambda \cdot b$$

and, since λ is positive, the proof is finished. □

Definition 1.14. In the first case of Proposition 1.13 we say that $\mathcal{E}$ and $\mathcal{F}$ have the *same orientation* and in the second case we say that $\mathcal{E}$ and $\mathcal{F}$ have *opposite orientation*.

Why is this relevant? Besides the fact that it gives a geometric interpretation of the sign of the determinant $\det(M_{\mathcal{E},\mathcal{F}})$, it also allows us to understand some signs which appear in calculations of areas (see Chapter ??). Moreover, the trigonometry that we know to hold true in $\mathbb{E}^2$ implicitly builds on the notion of oriented Euclidean plane $\mathbb{E}^2$. Mathematically, the distinction in orientation is only a matter of keeping track of the signs of some determinants. However, in relation to the physical world this distinction is more concrete.

Definition 1.15. Let $(\mathbf{i}, \mathbf{j})$ be a basis of $\mathbb{V}^2$ represented in a common point $O \in \mathbb{E}^2$ such that $\mathbf{i} = \overrightarrow{OX}$ and $\mathbf{j} = \overrightarrow{OY}$. Rotate the plane such that $\mathbf{i}$ points downward. If Y is in the right half-plane determined by the line OX , then we say that the basis $(\mathbf{i}, \mathbf{j})$ is *right oriented*. If Y lies in the left half-plane, we say that the basis $(\mathbf{i}, \mathbf{j})$ is *left oriented*. A coordinate system $(O, \mathbf{i}, \mathbf{j})$ is *left* or *right oriented* if the basis $(\mathbf{i}, \mathbf{j})$ is left or right oriented, respectively.

Fixing an orientation in the Euclidean plane $\mathbb{E}^2$ is equivalent to choosing a coordinate system $\mathcal{K} = (O, \mathcal{B})$ and calling it right oriented. Then, all other bases of $\mathbb{V}^2$ either have the same orientation as $\mathcal{B}$, in which case they are also called right oriented, or they have opposite orientation, in which case they are called left oriented. When it comes to a concrete configuration of points, on a sheet of paper for instance, such a choice can be made with the right-hand rule. Once we have such a choice, $\mathbb{E}^2$ is called oriented. In other words, the *oriented plane* $\mathbb{E}^2$ is the usual Euclidean plane together with a choice of which of the two opposite classes of bases contains the “preferred” bases.

Definition 1.16. Two bases $\mathcal{E}$ and $\mathcal{F}$ of the vector space $D(\mathbb{E}^n)$ are said to have the *same orientation* if $\det(M_{\mathcal{E}, \mathcal{F}}) > 0$. They have *opposite orientation* if $\det(M_{\mathcal{E}, \mathcal{F}}) < 0$. Two frames $\mathcal{K}$ and $\mathcal{K}'$ have the *same orientation* if their bases have the same orientation, and they have *opposite orientation* otherwise.

We say that the Euclidean space $\mathbb{E}^n$ is *oriented* if there is a choice of a frame $\mathcal{K} = (O, \mathcal{B})$ which is called *right oriented*. Then, all other frames of $\mathbb{E}^n$ with the same orientation as $\mathcal{K}$ are also *right oriented* and all other bases with opposite orientation are called *left oriented*.

Remark. Unless otherwise stated, whenever we consider a frame $\mathcal{K} = (O, \mathcal{B})$ of $\mathbb{E}^n$, we will assume that it is right oriented and that $\mathbb{E}^n$ is therefore an oriented Euclidean space.

CHAPTER 2

Affine subspaces

Contents

2.1 Lines in $\mathbb{A}^2$	17
2.1.1 Parametric equations	17
2.1.2 Cartesian equations	18
2.1.3 Relative positions of two lines in $\mathbb{A}^2$	19
2.2 Planes in $\mathbb{A}^3$	20
2.2.1 Parametric equations	20
2.2.2 Cartesian equations	21
2.2.3 Relative positions of two planes in $\mathbb{A}^3$	22
2.3 Lines in $\mathbb{A}^3$	23
2.3.1 Parametric equations	23
2.3.2 Cartesian equations	24
2.3.3 Relative positions of two lines in $\mathbb{A}^3$	24
2.3.4 Relative positions of a line and a plane in $\mathbb{A}^3$	25
2.4 Affine subspaces of $\mathbb{A}^n$	26
2.4.1 Hyperplanes	28
2.4.2 Lines	29
2.4.3 Relative positions	30
2.4.4 Changing the reference frame	32

2.1 Lines in $\mathbb{A}^2$

The Euclidean plane $\mathbb{E}^2$ is a 2-dimensional real affine space. In order to emphasize the fact that we treat $\mathbb{E}^2$ as an affine space only, we denote it by $\mathbb{A}^2$. In this setting, a line in $\mathbb{A}^2$ is an affine subspace of dimension 1, i.e., $D(S)$ is a 1-dimensional vector subspace of $\mathbb{V}^2 = D(\mathbb{A}^2)$ (see Definition 1.4). In terms of the map $\mathbf{r}_Q : \mathbb{A}^2 \rightarrow \mathbb{V}^2$, which identifies points P with position vectors relative to a fixed point $Q \in \mathbb{A}^2$, the subset $S \subseteq \mathbb{A}^2$ is a line if and only if for a point $Q \in S$ we have

$$\mathbf{r}_Q(S) = \{\overrightarrow{QP} : P \in S\} \text{ is a 1-dimensional vector subspace of } \mathbb{V}^2.$$

It is not difficult to see that if the above description holds for one point $Q \in S$, it holds for any point $Q \in S$. The nonzero elements of the direction space $D(S)$ of the line S are called *direction vectors*.

2.1.1 Parametric equations

For any two distinct points P, Q on a line S , the vector $\overrightarrow{QP}$ is a direction vector of S . Since $D(S)$ is 1-dimensional, all direction vectors are linearly dependent, i.e., $\mathbf{v}$ is a direction vector for S if and only if it is linearly dependent on $\overrightarrow{QP}$. So, for any direction vector $\mathbf{v}$ of S there is a *unique* scalar $t \in \mathbb{R}$ such that

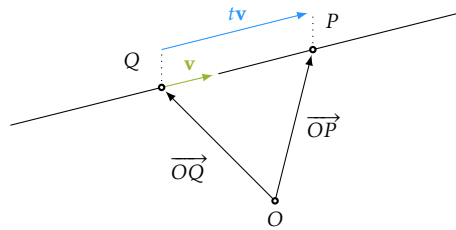
$$\overrightarrow{QP} = t\mathbf{v}.$$

Now, if we fix Q and let P vary on the line, then t varies in $\mathbb{R}$. Since $\mathbf{r}_Q : \mathbb{A}^2 \rightarrow \mathbb{V}^2$ is a bijection, the line S can be described as

$$S = \left\{ P \in \mathbb{A}^2 : \overrightarrow{QP} = t\mathbf{v} \text{ for some } t \in \mathbb{R} \right\}.$$

In this description, the point $Q \in S$ is arbitrary but fixed. If we want to emphasize that the description depends on fixing Q , we refer to this point as the *base point (of this description)* of the line. Moreover, for any point $O \in \mathbb{A}^2$ we may split the vector $\overrightarrow{QP}$ in the equation $\overrightarrow{QP} = t\mathbf{v}$ to obtain

$$\overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v}. \quad (2.1)$$



So, we can describe the line S as the set of points P in $\mathbb{A}^2$ which satisfy Equation (2.1) for some $t \in \mathbb{R}$. This equation is called the *vector equation of the line S relative to O , having base point Q and direction vector $\mathbf{v}$* , or simply a *vector equation* of the line S . If $\mathbf{v} = \overrightarrow{QA}$, this equation also describes the segment $[QA]$ if we restrict the parameter to $t \in [0, 1]$. Moreover, for $t > 0$ and $t < 0$ we obtain two rays emanating from Q .

Notice that a vector equation depends on the choice of the base point Q and on the choice of the direction vector $\mathbf{v}$. In particular, a line does not have a unique vector equation. Notice also that the vector equation does not depend on the frame. In the above description O can be any point in $\mathbb{A}^2$.

Now fix a frame $\mathcal{K} = (O, \mathcal{B})$. If we write Equation (2.1) in coordinates relative to $\mathcal{K}$, we obtain

$$S : \begin{cases} x = x_Q + tv_x \\ y = y_Q + tv_y \end{cases} \quad \text{or, in matrix form} \quad S : \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \end{bmatrix} \quad (2.2)$$

where $Q = Q(x_Q, y_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y)$ and where t is the *parameter*; for different values of t we obtain different points (x, y) on the line. The two equations in the system (2.2) are called *parametric equations* of the line S . Traditionally, they are written in the form of a system of equations as indicated on the left. Writing them as one equation, as indicated on the right, is closer to the computational perspective where we identify points with column matrices. Clearly, the two ways of writing such parametric equations are equivalent.

2.1.2 Cartesian equations

It is possible to eliminate the parameter t in (2.2). By expressing t in both equations and setting the two expressions equal, we obtain

$$\frac{x - x_Q}{v_x} = \frac{y - y_Q}{v_y}. \quad (2.3)$$

We refer to Equation (2.3) as a *symmetric equation* of the line S . It could happen that v_x or v_y is zero. In that case, translate back to the parametric equations to understand what happens.

Example 2.1. The line with symmetric equation

$$\frac{x - 3}{2} = \frac{y - 5}{0} \quad \text{has parametric equations} \quad \begin{cases} x = 3 + 2 \cdot t \\ y = 5 + 0 \cdot t \end{cases}.$$

Thus, it is the line parallel to Ox described by the equation $y = 5$.

We have just described a line with a linear equation relative to the frame $\mathcal{K}$. The converse is also true.

Proposition 2.2. Every line in $\mathbb{A}^2$ can be described with a linear equation in two variables

$$ax + by + c = 0 \quad (2.4)$$

relative to a fixed frame, and any linear equation in two variables relative to a fixed frame describes a line if the constants a and b are not both zero. Moreover, if Equation (2.4) describes the line ℓ relative to a frame $\mathcal{K} = (O, \mathcal{B})$, then the direction space $D(\ell)$ of the line is the 1-dimensional subspace of $\mathbb{V}^2$ which, relative to the basis $\mathcal{B}$, satisfies the equation

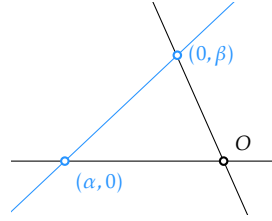
$$D(\ell) : ax + by = 0.$$

Proof. This is a particular case of Theorem 2.6. □

Equation (2.4) is called a *Cartesian equation* of the line which it describes. Notice that there are infinitely many Cartesian equations describing the same line, since we can multiply one equation by a nonzero constant. It is sometimes useful to rearrange the linear equation (2.4) in order to emphasize some geometric properties. For example, we can rearrange it in the form

$$\frac{x}{\alpha} + \frac{y}{\beta} = 1 \quad \text{where} \quad \alpha = -\frac{c}{a} \quad \text{and} \quad \beta = -\frac{c}{b}.$$

In this form, we have the *equation of the line where we can read off the intersection points with the coordinate axes* since this line intersects Ox in $(\alpha, 0)$ and it intersects Oy in $(0, \beta)$.



Alternatively, we may express the linear equation (2.4) with a determinant

$$\begin{vmatrix} x - x_Q & y - y_Q \\ v_x & v_y \end{vmatrix} = 0$$

which says that a point P belongs to the line if $\overrightarrow{QP}$ is linearly dependent on $\mathbf{v}$. In this form, we may describe the two half-planes which are separated by the line with the inequalities

$$\begin{vmatrix} x - x_Q & y - y_Q \\ v_x & v_y \end{vmatrix} < 0 \quad \text{and} \quad \begin{vmatrix} x - x_Q & y - y_Q \\ v_x & v_y \end{vmatrix} > 0$$

Indeed, any point P in the plane either lies on the line or $(\overrightarrow{QP}, \mathbf{v})$ is a left-oriented basis or $(\overrightarrow{QP}, \mathbf{v})$ is a right-oriented basis.

2.1.3 Relative positions of two lines in $\mathbb{A}^2$

The tools of linear algebra readily apply to describe intersections of lines in $\mathbb{A}^2$. Assume that we have two lines

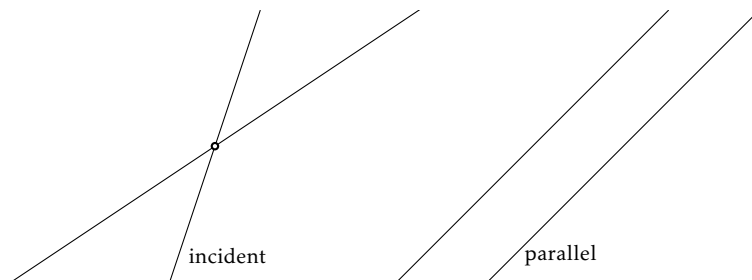
$$\ell_1 : a_1x + b_1y + c_1 = 0 \quad \text{and} \quad \ell_2 : a_2x + b_2y + c_2 = 0.$$

In order to determine if they intersect, one has to discuss the system:

$$\begin{cases} \ell_1 : a_1x + b_1y + c_1 = 0 \\ \ell_2 : a_2x + b_2y + c_2 = 0 \end{cases} . \quad (2.5)$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). In the plane, the situation is very simple:

- two lines intersect in a unique point, the coordinates of which are the solution to (2.5); or
- they don't intersect and (2.5) doesn't have solutions, in which case the lines are parallel; or
- the system (2.5) has infinitely many solutions, in which case $\ell_1 = \ell_2$.



2.2 Planes in $\mathbb{A}^3$

The usual Euclidean space $\mathbb{E}^3$ is a 3-dimensional real affine space. In order to emphasize the fact that we treat $\mathbb{E}^3$ as an affine space only, we denote it by $\mathbb{A}^3$. A plane in $\mathbb{A}^3$ is a set of points S which is an affine subspace of dimension 2 (see Definition 1.4). Considering the bijection $\mathbf{r}_Q : \mathbb{A}^3 \rightarrow \mathbb{V}^3$ which identifies points with position vectors relative to a fixed point $Q \in \mathbb{A}^3$, the subset S is a plane if and only if for any $Q \in S$

$$\mathbf{r}_Q(S) = \{\overrightarrow{QP} : P \in S\} \text{ is a 2-dimensional vector subspace of } \mathbb{V}^3.$$

It is not difficult to see that the above description does not depend on the point $Q \in S$. If S is a plane, the direction space $D(S) = \mathbf{r}_Q(S)$ is a 2-dimensional vector subspace of $\mathbb{V}^3$.

2.2.1 Parametric equations

Since $D(S)$ is a 2-dimensional vector space, any basis will contain two vectors. Let $(\mathbf{v}, \mathbf{w})$ be a basis of $D(S)$. Then, for any two points $P, Q \in S$, the vector $\overrightarrow{QP}$ is a linear combination of the basis vectors, i.e., there exist unique scalars $s, t \in \mathbb{R}$ such that

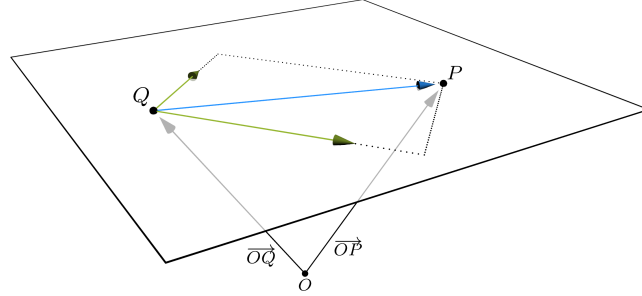
$$\overrightarrow{QP} = s\mathbf{v} + t\mathbf{w}.$$

Now, if we fix Q and let P vary in the plane S , then s and t vary in $\mathbb{R}$. Since $\mathbf{r}_Q : \mathbb{A}^3 \rightarrow \mathbb{V}^3$ is a bijection, the plane S can be described as

$$S = \left\{ P \in \mathbb{A}^3 : \overrightarrow{QP} = s\mathbf{v} + t\mathbf{w} \text{ for some } s, t \in \mathbb{R} \right\}.$$

In this description, the point $Q \in S$ is arbitrary but fixed. If we want to emphasize that the description depends on fixing Q , we refer to this point as the *base point (of this description)*. Moreover, for any point $O \in \mathbb{A}^3$ we may split the vector $\overrightarrow{QP}$ in the equation $\overrightarrow{QP} = s\mathbf{v} + t\mathbf{w}$ to obtain

$$\overrightarrow{OP} = \overrightarrow{OQ} + s\mathbf{v} + t\mathbf{w}. \quad (2.6)$$



So, we can describe the plane S as the set of points P in $\mathbb{A}^3$ which satisfy Equation (2.6) for some $s, t \in \mathbb{R}$. This equation is called the *vector equation of the plane S relative to O , having base point Q and direction vectors $\mathbf{v}$ and $\mathbf{w}$* , or simply a *vector equation* of the plane S . If $\mathbf{v} = \overrightarrow{QA}$, $\mathbf{w} = \overrightarrow{QC}$, and $\mathbf{v} + \mathbf{w} = \overrightarrow{QB}$, this equation also describes the interior of the parallelogram $QACB$ if we restrict the parameters s, t to $(0, 1)$. The interior of the triangle QAC is obtained if we further impose the condition that $s + t < 1$. Moreover, for $t > 0$ and $t < 0$ we obtain two half-planes separated by the line QA .

Notice that a vector equation depends on the choice of the base point Q and on the choice of the vectors $\mathbf{v}$ and $\mathbf{w}$. In analogy with the case of the line in $\mathbb{A}^2$, we may call such vectors *direction vectors* for the plane S . In particular, a plane does not have a unique vector equation. Notice also that the vector equation does not depend on the frame. In the above description O can be any point in $\mathbb{A}^3$.

Now fix a frame $\mathcal{K} = (O, \mathcal{B})$. If we write Equation (2.6) in coordinates relative to $\mathcal{K}$, we obtain

$$S : \begin{cases} x = x_Q + sv_x + tw_x \\ y = y_Q + sv_y + tw_y \\ z = z_Q + sv_z + tw_z \end{cases} \quad \text{or, in matrix form} \quad S : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} + s \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} + t \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \quad (2.7)$$

where $Q = Q(x_Q, y_Q, z_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y, v_z)$, and $\mathbf{w} = \mathbf{w}(w_x, w_y, w_z)$. The values s and t are called *parameters*, and for different parameters we obtain different points (x, y, z) in the plane S . The three equations in the system (2.7) are called *parametric equations* for the plane S .

2.2.2 Cartesian equations

Eliminating the parameters s, t in (2.7) yields a linear equation in x, y, z . A more direct approach is to observe that a point $P(x, y, z)$ lies in the plane S if and only if the vector $\overrightarrow{QP}$ is linearly dependent on the vectors $\mathbf{v}$ and $\mathbf{w}$. This is equivalent to:

$$\begin{vmatrix} x - x_Q & y - y_Q & z - z_Q \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = 0. \quad (2.8)$$

In particular, four points Q, E, F , and G are coplanar if and only if the vectors $\overrightarrow{QE}$, $\overrightarrow{QF}$, and $\overrightarrow{QG}$ are linearly dependent, which is equivalent to

$$\begin{vmatrix} x_E - x_Q & y_E - y_Q & z_E - z_Q \\ x_F - x_Q & y_F - y_Q & z_F - z_Q \\ x_G - x_Q & y_G - y_Q & z_G - z_Q \end{vmatrix} = 0. \quad (2.9)$$

Notice that Equation (2.9) is just a restatement of the fact that four points Q, E, F , and G are coplanar if and only if the vectors $\overrightarrow{QE}$, $\overrightarrow{QF}$, and $\overrightarrow{QG}$ are linearly dependent. Notice also that if we replace the equals sign in (2.8) with inequalities, we describe the two half-spaces separated by this plane since, for a point P not in the given plane, $(\overrightarrow{QP}, \mathbf{v}, \mathbf{w})$ is a basis which is either left- or right-oriented.

We have just described a plane with a linear equation (Equation (2.8)) relative to the frame $\mathcal{K}$. The converse is also true.

Proposition 2.3. Every plane in $\mathbb{A}^3$ can be described with a linear equation in three variables

$$ax + by + cz + d = 0 \quad (2.10)$$

relative to a fixed frame, and any linear equation relative to a fixed frame in three variables describes a plane if the constants a, b , and c are not all zero. Moreover, if Equation (2.10) describes the plane π relative to a frame $\mathcal{K} = (O, \mathcal{B})$, then the direction space $D(\pi)$ of the plane is the 2-dimensional subspace of $\mathbb{V}^3$ which, relative to the basis $\mathcal{B}$, satisfies the equation

$$D(\pi) : ax + by + cz = 0.$$

Proof. This is a particular case of Theorem 2.6. □

Equation (2.10) is called a *Cartesian equation* of the plane it describes. Notice that there are infinitely many Cartesian equations describing the same plane, since we can multiply one equation by a nonzero constant. Here again, it may be useful to rearrange the linear equation (2.10) in order to emphasize some geometric properties. For example, we can rearrange it in the form

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1 \quad \text{where} \quad \alpha = -\frac{d}{a}, \quad \beta = -\frac{d}{b} \quad \text{and} \quad \gamma = -\frac{d}{c}.$$

In this form, we have the *equation of the plane where we can read off the intersection points with the coordinate axes* since the plane intersects Ox in $(\alpha, 0, 0)$, it intersects Oy in $(0, \beta, 0)$, and it intersects Oz in $(0, 0, \gamma)$.

2.2.3 Relative positions of two planes in $\mathbb{A}^3$

In order to describe intersections of planes, we make use of linear algebra. Assume that we have two planes

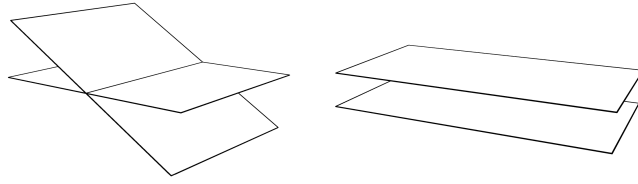
$$\pi_1 : a_1x + b_1y + c_1z + d_1 = 0 \quad \text{and} \quad \pi_2 : a_2x + b_2y + c_2z + d_2 = 0.$$

We determine if they intersect or not by discussing the system:

$$\begin{cases} \pi_1 : a_1x + b_1y + c_1z + d_1 = 0 \\ \pi_2 : a_2x + b_2y + c_2z + d_2 = 0 \end{cases} \quad (2.11)$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). Here again, the situation is very simple. Let M be the matrix of the system and $\tilde{M}$ the extended matrix of the system. Then we have:

- two planes either intersect in a line (the coordinates of the points on the line will be solutions to (2.11)); this happens if the rank of M and the rank of $\tilde{M}$ equal 2; or
- they don't intersect and (2.11) doesn't have solutions, in which case the planes are parallel; this happens if the rank of M is strictly less than the rank of $\tilde{M}$; or
- the solutions to system (2.11) depend on two parameters, in which case $\pi_1 = \pi_2$; this happens if the rank of M and the rank of $\tilde{M}$ are equal to 1.



2.3 Lines in $\mathbb{A}^3$

Here again, we treat the usual Euclidean space $\mathbb{E}^3$ as a 3-dimensional affine space and denote it by $\mathbb{A}^3$. As in the case of $\mathbb{A}^2$, by Definition 1.4, a line in $\mathbb{A}^3$ is an affine subspace of dimension 1. Hence, the subset S is a line if for any $Q \in S$ we have that

$$\mathbf{r}_Q(S) = \{\overrightarrow{QP} : P \in S\} \text{ is a 1-dimensional vector subspace of } \mathbb{V}^3.$$

2.3.1 Parametric equations

The vector description of a line in $\mathbb{A}^3$ is identical to that in $\mathbb{A}^2$. Given a base point Q and a direction vector $\mathbf{v}$, the line S is the set of points P satisfying:

$$\overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v}, \quad t \in \mathbb{R}. \quad (2.12)$$

Fixing a frame $\mathcal{K} = (O, \mathcal{B})$, we write Equation (2.12) in coordinates to obtain:

$$\begin{cases} x = x_Q + tv_x \\ y = y_Q + tv_y \\ z = z_Q + tv_z \end{cases} \quad \text{or, in matrix form} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_Q \\ y_Q \\ z_Q \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \quad (2.13)$$

where $Q = Q(x_Q, y_Q, z_Q)$, $\mathbf{v} = \mathbf{v}(v_x, v_y, v_z)$ relative to $\mathcal{K}$, and where t is the *parameter* yielding different points (x, y, z) on the line. The three equations in the system (2.13) are called *parametric equations* for the line S .

2.3.2 Cartesian equations

It is possible to eliminate the parameter t in (2.13) in order to obtain

$$\frac{x - x_Q}{v_x} = \frac{y - y_Q}{v_y} = \frac{z - z_Q}{v_z}. \quad (2.14)$$

We refer to Equation (2.14) as *symmetric equations* of the line S . It could happen that v_x , v_y , or v_z is zero. In that case, translate back to the parametric equations to understand what happens.

We have just described a line with two linear equations (Equations (2.14)) relative to the frame $\mathcal{K}$. The converse is also true.

Proposition 2.4. Every line in $\mathbb{A}^3$ can be described with two linear equations in three variables

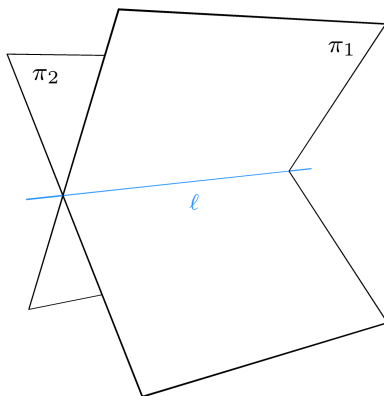
$$\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases} \quad (2.15)$$

relative to a fixed frame, and any compatible system of two linear equations of rank 2 in three variables relative to a fixed frame describes a line. Moreover, if the system of equations (2.15) describes the line ℓ relative to a frame $\mathcal{K} = (O, \mathcal{B})$, then the direction space $D(\ell)$ of the line is the 1-dimensional subspace of $\mathbb{V}^3$ which, relative to the basis $\mathcal{B}$, satisfies the equations

$$D(\ell) : \begin{cases} a_1x + b_1y + c_1z = 0 \\ a_2x + b_2y + c_2z = 0 \end{cases}. \quad (2.16)$$

Proof. This is a particular case of Theorem 2.6. □

The system of equations (2.15) is called a system of *Cartesian equations* of the line which it describes. Notice that it describes a line as an intersection of two planes.



2.3.3 Relative positions of two lines in $\mathbb{A}^3$

Again, the intersections of lines can be determined with linear algebra. Assume we have two lines

$$\ell_1 : \begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases} \quad \text{and} \quad \ell_2 : \begin{cases} a_3x + b_3y + c_3z + d_3 = 0 \\ a_4x + b_4y + c_4z + d_4 = 0 \end{cases}.$$

One way to determine if they intersect is to discuss the system:

$$\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \\ a_3x + b_3y + c_3z + d_3 = 0 \\ a_4x + b_4y + c_4z + d_4 = 0 \end{cases} . \quad (2.17)$$

Discussing this system is basic linear algebra (see, for example, [9, Section 3.6]). It is somewhat easier to discuss the relative positions of lines in $\mathbb{A}^3$ via their parametric equations:

$$\ell_1 : \begin{cases} x = x_1 + tv_x \\ y = y_1 + tv_y \\ z = z_1 + tv_z \end{cases} \quad \text{and} \quad \ell_2 : \begin{cases} x = x_2 + tu_x \\ y = y_2 + tu_y \\ z = z_2 + tu_z \end{cases} .$$

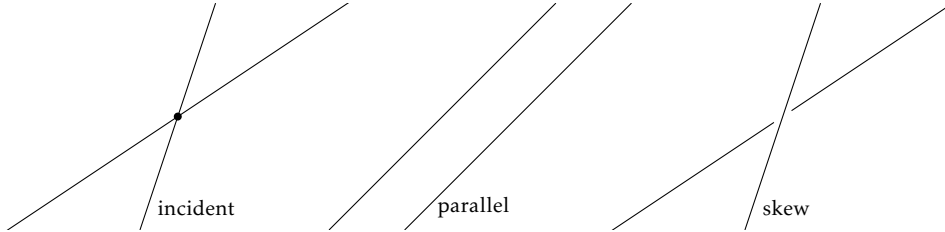
We have the following cases:

- if the direction vectors $\mathbf{v}(v_1, v_2, v_3)$ and $\mathbf{u}(u_1, u_2, u_3)$ are proportional, then the two lines are parallel;
- if they are parallel and have a point in common, then the two lines are equal;
- if they are not parallel, then they are coplanar (they lie in the same plane) if

$$\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ v_x & v_y & v_z \\ u_x & u_y & u_z \end{vmatrix} = 0,$$

in which case they intersect in exactly one point;

- if they are not parallel and they don't intersect, then we say that the two lines ℓ_1 and ℓ_2 are *skew* relative to each other.



2.3.4 Relative positions of a line and a plane in $\mathbb{A}^3$

Consider the plane

$$\pi : ax + by + cz + d = 0$$

and the line

$$\ell : \begin{cases} x = x_0 + tv_x \\ y = y_0 + tv_y \\ z = z_0 + tv_z \end{cases} .$$

In order to see if they intersect, we check to see which points in ℓ satisfy the equation of π :

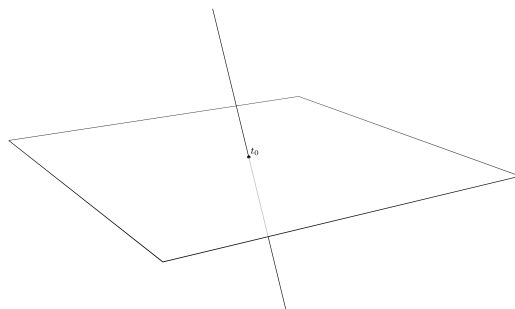
$$a(x_0 + tv_x) + b(y_0 + tv_y) + c(z_0 + tv_z) + d = 0 \Leftrightarrow (av_x + bv_y + cv_z)t + ax_0 + by_0 + cz_0 + d = 0. \quad (2.18)$$

The possibilities are:

- $av_x + bv_y + cv_z = 0$ and $ax_0 + by_0 + cz_0 + d \neq 0$, in which case Equation (2.18) has no solution, i.e., the plane and the line don't intersect (they are parallel); or
- $av_x + bv_y + cv_z = 0$ and $ax_0 + by_0 + cz_0 + d = 0$, in which case any $t \in \mathbb{R}$ is a solution to Equation (2.18), i.e., the line is contained in the plane (in particular, they are parallel); or
- $av_x + bv_y + cv_z \neq 0$, in which case Equation (2.18) has the unique solution

$$t_0 = -\frac{ax_0 + by_0 + cz_0 + d}{av_x + bv_y + cv_z}.$$

Hence, the point corresponding to the parameter t_0 is the intersection point $\ell \cap \pi$.



2.4 Affine subspaces of $\mathbb{A}^n$

Recall Definition 1.4 for affine subspaces. Two affine subspaces $S_1, S_2 \subseteq \mathbb{A}^n$ are *parallel* ($S_1 \parallel S_2$) if $D(S_1) \subseteq D(S_2)$ or $D(S_2) \subseteq D(S_1)$.

Affine subspaces are affine spaces in their own right, inheriting the structure from $\mathbb{A}^n$, analogous to vector subspaces.

Proposition 2.5. An affine subspace of $\mathbb{A}^n$ is an affine space with the affine structure inherited from $\mathbb{A}^n$.

Proof. The proof is a simple matter of unpacking the definition of affine spaces (Definition 1.2). The space $\mathbb{A}^n$ is a triple $(\mathbb{P}, \mathbb{V}, t)$, where $\mathbb{P}$ is the set of points, $\mathbb{V}$ is an n -dimensional vector space, and $t : \mathbb{V} \times \mathbb{P} \rightarrow \mathbb{P}$ is the translation map. If S is an affine subspace, it is in particular a subset of $\mathbb{P}$. The direction space $D(S)$ consists of vectors in $\mathbb{V}$ which can be represented by points in S , thus $D(S)$ is a vector subspace of $\mathbb{V}$. The inherited affine structure is obtained by restricting $\square - \square : S \times S \rightarrow D(S)$ and $\square + \square : D(S) \times S \rightarrow S$. Since these maps satisfy the axioms of an affine space, so do their restrictions. Hence the triple $(S, D(S), t')$ is an affine space. $\square$

Fixing a point $O \in \mathbb{A}^n$, a point $Q \in S$, and a basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d)$ of $D(S)$, it follows from the definition that S has the following description:

$$S = \left\{ P \in \mathbb{A}^n : \overrightarrow{OP} = \overrightarrow{OQ} + t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + \dots + t_d \mathbf{v}_d \text{ for some } t_1, \dots, t_d \in \mathbb{R} \right\}. \quad (2.19)$$

The equation in (2.19) is called the *vector equation of the affine subspace S relative to O , having base point Q and direction vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d$* , or simply a *vector equation* of S .

Fixing a frame with origin O and translating the equation in (2.19) into coordinates, one obtains *parametric equations* of the affine space S .

$$S : \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t_1 \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \dots + t_d \begin{bmatrix} v_{d,1} \\ v_{d,2} \\ \vdots \\ v_{d,n} \end{bmatrix} \quad t_1, \dots, t_d \in \mathbb{R}. \quad (2.20)$$

Another way of representing an affine subspace is by *Cartesian equations* (Equations (2.21)) as follows.

Theorem 2.6. Fix a frame $\mathcal{K} = (O, \mathcal{B})$ in the affine space $\mathbb{A}^n$. Let

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tn}x_n = b_t \end{cases} \quad (2.21)$$

be a system of linear equations in the unknowns $x_1, \dots, x_n$. The set S of points of $\mathbb{A}^n$ whose coordinates are solutions to (2.21), if there are any, is an affine space of dimension $d = n - r$, where r is the rank of the matrix of coefficients of the system. The direction space $D(S)$ is the vector subspace of $\mathbb{V}^n$ whose equations relative to $\mathcal{B}$ are given by the associated homogeneous system

$$D(S) : \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tn}x_n = 0 \end{cases}. \quad (2.22)$$

Conversely, for every affine subspace S of $\mathbb{A}^n$ of dimension d there is a system of $n - d$ linear equations in n unknowns whose solutions correspond precisely to the coordinates of the points in S .

Proof. We follow the proof in [20, Section 8]. Denote by $\mathbb{W}$ the vector subspace defined by the homogeneous system (2.22). First, we show that the set of solutions to (2.21) is an affine subspace of $\mathbb{A}^n$ with the indicated properties. By assumption, we only consider the cases where $S \neq \emptyset$. Thus, we may fix a point $Q(q_1, \dots, q_n) \in S$. Then, for any point $P(p_1, \dots, p_n)$ belonging to S , we have

$$a_{j1}(p_1 - q_1) + \dots + a_{jn}(p_n - q_n) = \underbrace{(a_{j1}p_1 + \dots + a_{jn}p_n)}_{b_j} - \underbrace{(a_{j1}q_1 + \dots + a_{jn}q_n)}_{b_j} = 0$$

for each $j = 1, \dots, t$. Thus, $\overrightarrow{QP} \in \mathbb{W}$. This shows that S is contained in the affine subspace T passing through Q and parallel to $\mathbb{W}$. Conversely, if $R(r_1, \dots, r_n) \in T$, then $\overrightarrow{QR} \in \mathbb{W}$ and so the components $(r_1 - q_1, \dots, r_n - q_n)$ of $\overrightarrow{QR}$ are solutions to (2.22). Thus,

$$0 = a_{j1}(r_1 - q_1) + \dots + a_{jn}(r_n - q_n) = a_{j1}r_1 + \dots + a_{jn}r_n - \underbrace{(a_{j1}q_1 + \dots + a_{jn}q_n)}_{b_j}$$

for each $j = 1, \dots, t$. That is, $R \in S$. Thus $S = T$, hence S is an affine subspace. Moreover, $\dim(S) = \dim(\mathbb{W}) = n$ minus the rank of the matrix of coefficients of (2.22).

Next, we show that an affine subspace $S \subseteq \mathbb{A}^n$ has a description by a linear system of the form (2.21). Let S be any affine subspace of $\mathbb{A}^n$ with direction space $\mathbb{W}$ of dimension s . Being an s -dimensional subspace of $\mathbb{V}$, $\mathbb{W}$ can be described by a homogeneous system with $n - s$ equations

$$\mathbb{W}: \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ \vdots \\ a_{n-s,1}x_1 + \dots + a_{n-s,n}x_n = 0 \end{cases}$$

Fix a point $Q \in S$. The points $P(p_1, \dots, p_n)$ of S are characterized by the condition that $\overrightarrow{QP} \in \mathbb{W}$, i.e.,

$$a_{j1}(p_1 - q_1) + \dots + a_{jn}(p_n - q_n) = 0$$

for each $j = 1, \dots, n - s$, or, equivalently,

$$a_{j1}p_1 + \dots + a_{jn}p_n = b_j$$

where we have put $b_j = a_{j1}q_1 + \dots + a_{jn}q_n$. Thus, the points $P(p_1, \dots, p_n) \in S$ are precisely those points in $\mathbb{A}^n$ whose coordinates satisfy the equations

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{n-s,1}x_1 + \dots + a_{n-s,n}x_n = b_{n-s} \end{cases}$$

□

2.4.1 Hyperplanes

Recall that affine subspaces in $\mathbb{A}^n$ which have dimension $n - 1$ are called *hyperplanes* (see Definition 1.4). Let H be a hyperplane and let $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1})$ be a basis of $D(H)$. With respect to a frame $\mathcal{K} = (O, \mathbf{e}_1, \dots, \mathbf{e}_n)$ of $\mathbb{A}^n$, parametric equations of H are of the form

$$H: \begin{cases} x_1 = q_1 + t_1 v_{1,1} + \dots + t_{n-1} v_{n-1,1} \\ x_2 = q_2 + t_1 v_{1,2} + \dots + t_{n-1} v_{n-1,2} \\ \vdots \\ x_n = q_n + t_1 v_{1,n} + \dots + t_{n-1} v_{n-1,n} \end{cases} \quad \text{or} \quad H: \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t_1 \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \dots + t_{n-1} \begin{bmatrix} v_{n-1,1} \\ v_{n-1,2} \\ \vdots \\ v_{n-1,n} \end{bmatrix} \quad (2.23)$$

where $\mathbf{v}_i = \mathbf{v}_i(v_{i,1}, \dots, v_{i,n})$, where $Q = Q(q_1, \dots, q_n)$ is a point in H , and where $t_i \in \mathbb{R}$ for each $i \in \{1, \dots, n-1\}$. These parametric equations express the fact that a point P belongs to H if and only if the vector $\overrightarrow{QP}$ is a linear combination of the basis vectors $\mathbf{v}_1, \dots, \mathbf{v}_{n-1}$, i.e., if and only if the vectors $\overrightarrow{QP}, \mathbf{v}_1, \dots, \mathbf{v}_{n-1}$ are linearly dependent. We can reformulate this as follows. A point $P(x_1, \dots, x_n)$ belongs to the hyperplane H if and only if

$$\begin{vmatrix} x_1 - q_1 & x_2 - q_2 & \dots & x_n - q_n \\ v_{1,1} & v_{1,2} & \dots & v_{1,n} \\ \vdots & \vdots & & \vdots \\ v_{n-1,1} & v_{n-1,2} & \dots & v_{n-1,n} \end{vmatrix} = 0. \quad (2.24)$$

This is a *Cartesian equation* of the hyperplane H .

2.4.2 Lines

A line in $\mathbb{A}^n$ is a 1-dimensional affine subspace (see Definition 1.4). If ℓ is such a line, then the vectors which can be represented by points in ℓ are linearly dependent. Any such nonzero vector $\mathbf{v}$ is called a *direction vector* of ℓ . Thus, ℓ can be described as

$$\ell = \left\{ P \in \mathbb{A}^n : \overrightarrow{OP} = \overrightarrow{OQ} + t\mathbf{v} \text{ for some } t \in \mathbb{R} \right\}$$

for any point $O \in \mathbb{A}^n$ and any point $Q \in \ell$. The image that goes with this description is the same as the one in dimension 2, but here we interpret it in the n -dimensional space $\mathbb{A}^n$. In coordinates, relative to a given frame $\mathcal{K}$ of $\mathbb{A}^n$, we obtain parametric equations for ℓ . They are of the form:

$$\ell : \begin{cases} x_1 = q_1 + tv_1 \\ x_2 = q_2 + tv_2 \\ \vdots \\ x_n = q_n + tv_n \end{cases} \quad \text{or, in matrix notation,} \quad \ell : \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + t \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

where $Q = Q(q_1, \dots, q_n)$ and $\mathbf{v} = \mathbf{v}(v_1, \dots, v_n)$ relative to $\mathcal{K}$. Here again, it is possible to eliminate the parameter t in order to obtain *symmetric equations* of the line ℓ :

$$\ell : \frac{x_1 - q_1}{v_1} = \frac{x_2 - q_2}{v_2} = \dots = \frac{x_n - q_n}{v_n}.$$

This is in fact a system of $(n-1)$ linear equations which we can rearrange to look like this:

$$\ell : \begin{cases} a_{1,1}x_1 + \dots + a_{1,n}x_n = b_1 \\ \vdots \\ a_{n-1,1}x_1 + \dots + a_{n-1,n}x_n = b_{n-1} \end{cases}.$$

Notice that the rank of this system is $n-1$ since $\dim(\ell) = 1$. Moreover, notice that each linear equation in the above system describes a hyperplane. So, this is saying that a line in $\mathbb{A}^n$ can be described by the intersection of $n-1$ hyperplanes.

2.4.3 Relative positions

Let S and T be two affine subspaces of $\mathbb{A}^n$. If $S \parallel T$, then they are disjoint or one is included in the other (see Proposition 2.7 below). Notice that if $\dim(S) = \dim(T)$, then S and T are parallel if and only if $D(S) = D(T)$. In particular, if S and T are lines, they are parallel if they have the same direction, i.e., any two of their direction vectors are proportional. Notice also that two hyperplanes are parallel if the coefficients of the unknowns in their equations are proportional.

Proposition 2.7. Let S and T be parallel affine subspaces of $\mathbb{A}^n$ with $\dim(S) \leq \dim(T)$.

- 1) If S and T have a point in common, then $S \subseteq T$.
- 2) If $\dim(S) = \dim(T)$, and S and T have a point in common, then $S = T$.

Proof. Fix $Q \in S \cap T$. Since S and T are parallel, without loss of generality we may assume that $D(S) \subseteq D(T)$. Then, we obtain 1) from

$$S = \{P \in \mathbb{A}^n : \overrightarrow{QP} \in D(S)\} \subseteq \{P \in \mathbb{A}^n : \overrightarrow{QP} \in D(T)\} = T.$$

We obtain 2) by noticing that $\dim(S) = \dim(T)$ implies equality in the above equation. Indeed, by definition $\dim(S) = \dim(T)$ means that $\dim(D(S)) = \dim(D(T))$, but then $D(S)$ is a vector subspace of $D(T)$ of maximal dimension, hence $D(S) = D(T)$. $\square$

As a consequence of Proposition 2.7, we obtain the following corollary which implies the ‘parallel postulate’ of Euclidean geometry (Axiom IV in [14]). The axioms of affine spaces therefore imply the validity of this postulate.

Corollary 2.8. If S is an affine subspace of $\mathbb{A}^n$ and Q is a point in $\mathbb{A}^n$, there is a unique affine subspace T of $\mathbb{A}^n$ which contains Q , is parallel to S , and has the same dimension as S .

Proof. Fix a point $Q \in \mathbb{A}^n$ and a point $P \in S$. To see that T exists, translate all points of S by $\overrightarrow{QP}$, i.e., consider

$$T = \overrightarrow{QP} + S = \overrightarrow{QP} + \{P' \in \mathbb{A}^n : \overrightarrow{PP'} \in D(S)\} = \{P' \in \mathbb{A}^n : \overrightarrow{QP'} \in D(S)\}$$

where for the last equality we use $\overrightarrow{QP} + \overrightarrow{PP'} = \overrightarrow{QP'}$. Then T is an affine subspace with $D(T) = D(S)$; in particular, it is parallel to S , and $\dim(T) = \dim(S)$. To see that T is unique, assume that T' is another affine subspace passing through Q which is parallel to S , and of the same dimension. By point 2) of Proposition 2.7 we see that T' has to equal T . $\square$

Definition 2.9. If two affine subspaces S and T of $\mathbb{A}^n$ are not parallel, they are said to be *skew* if they do not meet, or *incident* if they have a point in common.

In order to determine the intersection $S \cap T$, suppose that the two subspaces are given by the Cartesian equations

$$S : \sum_{j=1}^n a_{ij}x_j = b_i \quad \text{for } i = 1, \dots, n-s \quad (2.25)$$

$$T : \sum_{j=1}^n c_{kj}x_j = d_k \quad \text{for } k = 1, \dots, n-t. \quad (2.26)$$

The intersection $S \cap T$ is the locus of points in $\mathbb{A}^n$ whose coordinates are simultaneously solutions to both (2.25) and (2.26), i.e., they are solutions to the system

$$S \cap T : \begin{cases} \sum_{j=1}^n a_{ij}x_j = b_i & \text{for } i = 1, \dots, n-s, \\ \sum_{j=1}^n c_{kj}x_j = d_k & \text{for } k = 1, \dots, n-t. \end{cases} \quad (2.27)$$

By Theorem 2.6, if the system (2.27) has a solution, then it describes an affine subspace. Thus, if $S \cap T$ is nonempty, it is an affine subspace of $\mathbb{A}^n$.

Proposition 2.10. If the intersection $S \cap T$ of two affine subspaces of $\mathbb{A}^n$ is nonempty, it is an affine subspace satisfying

$$\dim(S) + \dim(T) - \dim(\mathbb{A}^n) \leq \dim(S \cap T) \leq \min\{\dim(S), \dim(T)\}. \quad (2.28)$$

Proof. Let $s = \dim(S)$ and let $t = \dim(T)$. By Theorem 2.6 we have

$$\begin{cases} S : \sum_{j=1}^n a_{ij}x_j = 0 & \text{for } i = 1, \dots, n-s \\ T : \sum_{j=1}^n c_{ij}x_j = 0 & \text{for } i = 1, \dots, n-t \end{cases}$$

Since $S \cap T$ is nonempty, the above system is compatible, and the dimension of $S \cap T$ is $n-r$, where r is the rank of the matrix of coefficients of this system. Notice that

$$r \leq n-s+n-t = 2n-(s+t)$$

Thus,

$$\dim(S \cap T) = n-r \geq n - [2n-(s+t)] = s+t-n = \dim(S) + \dim(T) - \dim(\mathbb{A}^n).$$

The last inequality is clear since $S \cap T \subseteq S$ and $S \cap T \subseteq T$ implies $D(S \cap T) \subseteq D(S)$ and $D(S \cap T) \subseteq D(T)$, and therefore $\dim(S \cap T) \leq \dim(S)$ and $\dim(S \cap T) \leq \dim(T)$. $\square$

Notice that in the previous proposition, the second inequality is an equality if $S \subseteq T$ or $T \subseteq S$. What about the first inequality? When do we have equality there?

Proposition 2.11. Let S and T be two affine subspaces of $\mathbb{A}^n$. Then $D(\mathbb{A}^n) = D(S) + D(T)$ if and only if $S \cap T \neq \emptyset$ and

$$\dim(S \cap T) = \dim(S) + \dim(T) - \dim(\mathbb{A}^n). \quad (2.29)$$

Proof. We use Grassmann's identity, which states that for any vector subspaces $\mathbb{W}$ and $\mathbb{U}$ we have

$$\dim(\mathbb{W}) + \dim(\mathbb{U}) = \dim(\mathbb{W} \cap \mathbb{U}) + \dim(\mathbb{W} + \mathbb{U}).$$

Let $s = \dim(S)$ and let $t = \dim(T)$. By Theorem 2.6 we have

$$\begin{cases} S : \sum_{j=1}^n a_{ij}x_j = b_i & \text{for } i = 1, \dots, n-s \\ T : \sum_{j=1}^n c_{kj}x_j = d_k & \text{for } k = 1, \dots, n-t \end{cases}$$

By convention, $\dim(\emptyset) = -\infty$; thus, (2.29) holds only if $S \cap T \neq \emptyset$. Assume that $S \cap T$ is nonempty. Then, the above system is compatible, and the dimension of $S \cap T$ is $n - r$, where r denotes the rank of the matrix of coefficients of this system. Moreover, we notice that (2.29) is equivalent to

$$n - r = s + t - n \Leftrightarrow \dim(D(S \cap T)) = \dim(D(S)) + \dim(D(T)) - \dim(\mathbb{V}^n)$$

and, since $S \cap T \neq \emptyset$, this is in turn equivalent to

$$\dim(D(S) \cap D(T)) = \dim(D(S)) + \dim(D(T)) - \dim(\mathbb{V}^n).$$

Rearranging the equation and using Grassmann's identity, the equation is equivalent to

$$\dim(\mathbb{V}^n) = \dim(D(S)) + \dim(D(T)) - \dim(D(S) \cap D(T)) = \dim(D(S) + D(T)).$$

At this point, we use the fact that the vector subspace $D(S) + D(T)$ of $\mathbb{V}^n$ has maximal dimension if and only if it equals the ambient space, i.e., if and only if $D(S) + D(T) = \mathbb{V}^n$. $\square$

2.4.4 Changing the reference frame

Let S be an affine subspace of $\mathbb{A}^n$ given with respect to the frame $\mathcal{K} = (O, \mathcal{B})$ via the parametric equations (2.20). Then, if $\mathcal{K}' = (O', \mathcal{B}')$ is another frame, by Theorem 1.10, the parametric equations with respect to $\mathcal{K}'$ are

$$S : \begin{bmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{bmatrix} = M_{\mathcal{B}', \mathcal{B}} \cdot \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + [O]_{\mathcal{K}'} + t_1 \cdot M_{\mathcal{B}', \mathcal{B}} \cdot \begin{bmatrix} v_{1,1} \\ v_{1,2} \\ \vdots \\ v_{1,n} \end{bmatrix} + \cdots + t_d \cdot M_{\mathcal{B}', \mathcal{B}} \cdot \begin{bmatrix} v_{d,1} \\ v_{d,2} \\ \vdots \\ v_{d,n} \end{bmatrix}.$$

In terms of Cartesian equations, notice that (2.21) can be written in the form

$$S : A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = b.$$

Then, with respect to $\mathcal{K}'$, this system translates as follows:

$$S : \underbrace{(A \cdot M_{\mathcal{B}, \mathcal{B}'})}_{=A'} \cdot \begin{bmatrix} x'_1 \\ \vdots \\ x'_n \end{bmatrix} = \underbrace{b - A \cdot [O']_{\mathcal{K}}}_{=b'}.$$

Changing the basis in a vector space

Let $\phi : V \rightarrow W$ be a linear map between the vector spaces V and W . Let $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be a basis for V and let $\mathcal{F}$ be a basis for W . In your algebra course, you used the notation $[\phi]_{\mathcal{E}, \mathcal{F}}$ for the matrix of the linear map ϕ with respect to the bases $\mathcal{E}$ and $\mathcal{F}$ [9, Definition 3.4.1]. We will use the notation

$$M_{\mathcal{F}, \mathcal{E}}(\phi) = [\phi]_{\mathcal{E}, \mathcal{F}}.$$

Notice that the indices $\mathcal{E}, \mathcal{F}$ are *reversed*. Recall that this is the matrix whose columns are the components of the $\phi(\mathbf{e}_i)$'s with respect to the basis $\mathcal{F}$:

$$M_{\mathcal{F}, \mathcal{E}}(\phi) = \begin{bmatrix} \uparrow & \dots & \uparrow \\ [\phi(\mathbf{e}_1)]_{\mathcal{F}} & \dots & [\phi(\mathbf{e}_n)]_{\mathcal{F}} \\ \downarrow & \dots & \downarrow \end{bmatrix}.$$

You have also learned [9, Theorem 3.4.8] that if $\psi : W \rightarrow U$ is another linear map to some vector space U with basis $\mathcal{G}$, then

$$M_{\mathcal{G}, \mathcal{E}}(\psi \circ \phi) = M_{\mathcal{G}, \mathcal{F}}(\psi) \cdot M_{\mathcal{F}, \mathcal{E}}(\phi).$$

In particular, if $V = W = U$, $\mathcal{G} = \mathcal{F}$, and $\phi = \psi = \text{Id}_V$, then

$$I_n = M_{\mathcal{F}, \mathcal{F}}(\text{Id}_V) = M_{\mathcal{F}, \mathcal{F}}(\text{Id}_V \circ \text{Id}_V) = M_{\mathcal{F}, \mathcal{E}}(\text{Id}_V) \cdot M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V)$$

hence, $M_{\mathcal{F}, \mathcal{E}}(\text{Id}_V) = M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V)^{-1}$, and thus

$$M_{\mathcal{F}, \mathcal{F}}(\phi) = M_{\mathcal{F}, \mathcal{E}}(\text{Id}_V) \cdot M_{\mathcal{E}, \mathcal{E}}(\phi) \cdot M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V) = M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V)^{-1} \cdot M_{\mathcal{E}, \mathcal{E}}(\phi) \cdot M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V).$$

So, the matrix of ϕ with respect to the basis $\mathcal{F}$ is obtained from the matrix of ϕ with respect to the basis $\mathcal{E}$ by conjugating with the matrix $M_{\mathcal{E}, \mathcal{F}}(\text{Id}_V)$.

Definition A.1. The matrix $M_{\mathcal{E},\mathcal{F}} := M_{\mathcal{E},\mathcal{F}}(\text{Id}_V)$ is called the *change of basis matrix from the basis $\mathcal{F}$ to the basis $\mathcal{E}$* . It is the matrix whose columns are the components of the vectors in $\mathcal{F}$ with respect to $\mathcal{E}$. If $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_n)$, then

$$M_{\mathcal{E},\mathcal{F}} = \begin{bmatrix} \uparrow & \dots & \uparrow \\ [\mathbf{f}_1]_{\mathcal{E}} & \dots & [\mathbf{f}_n]_{\mathcal{E}} \\ \downarrow & \dots & \downarrow \end{bmatrix}.$$

Moreover, we have $M_{\mathcal{F},\mathcal{E}} = (M_{\mathcal{E},\mathcal{F}})^{-1}$.

Proposition A.2. Let $\mathbf{v} \in V$. The coordinates of $\mathbf{v}$ with respect to the basis $\mathcal{E}$ are obtained from the coordinates of $\mathbf{v}$ with respect to the basis $\mathcal{F}$ by multiplication with the change of basis matrix:

$$[\mathbf{v}]_{\mathcal{E}} = M_{\mathcal{E},\mathcal{F}} \cdot [\mathbf{v}]_{\mathcal{F}}.$$

B.1 Polar coordinates

Polar coordinates use oriented angles with a line. They establish a bijection between points distinct from the origin and pairs of numbers (r, φ) in $\mathbb{R}_{>0} \times [0, 2\pi)$.

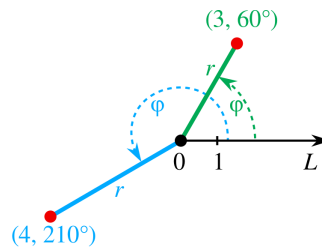


Figure B.1: Polar coordinates¹.

The correspondence between Cartesian coordinates (x, y) and polar coordinates (r, φ) is as follows:

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}, \quad \begin{cases} r = r(x, y) = \sqrt{x^2 + y^2} \\ \varphi = \varphi(x, y) \end{cases}$$

where, with $r \neq 0$,

$$\varphi(x, y) = \begin{cases} \arccos(\frac{x}{r}) & \text{if } y \geq 0 \\ -\arccos(\frac{x}{r}) & \text{if } y < 0 \end{cases}.$$

¹Image source: Wikipedia

B.2 Cylindrical coordinates

Cylindrical coordinates extend polar coordinates to dimension 3. They establish a bijection between points distinct from the origin and triples of numbers (r, φ, z) in $\mathbb{R}_{>0} \times [0, 2\pi) \times \mathbb{R}$.

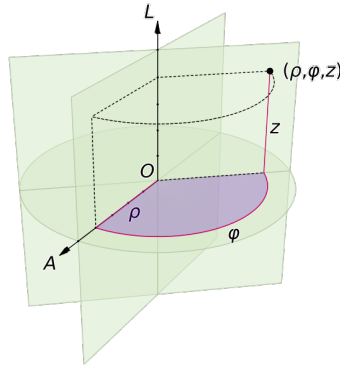


Figure B.2: Cylindrical coordinates².

The correspondence between Cartesian coordinates (x, y, z) and cylindrical coordinates (r, φ, z) is as follows:

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = z \end{cases}, \quad \begin{cases} r = r(x, y) = \sqrt{x^2 + y^2} \\ \varphi = f(x, y) \\ z = z \end{cases}$$

where f is as in Section B.1.

B.3 Spherical coordinates

Spherical coordinates also extend polar coordinates to dimension 3 but use angles for the third coordinate. They establish a bijection between points distinct from the origin and triples of numbers (r, φ, θ) in $\mathbb{R}_{>0} \times [0, 2\pi) \times [0, \pi)$.

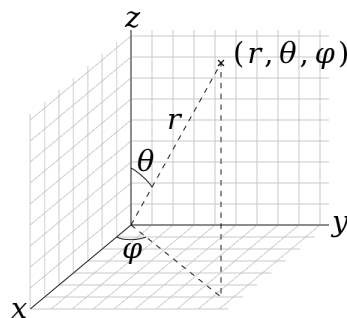


Figure B.3: Spherical coordinates³.

²Image source: Wikipedia

The correspondence between Cartesian coordinates (x, y, z) and spherical coordinates (r, φ, θ) is as follows:

$$\begin{cases} x = r \cos \varphi \sin \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \theta \end{cases}, \quad \begin{cases} r = r(x, y, z) = \sqrt{x^2 + y^2 + z^2} \\ \varphi = \varphi(x, y) \\ \theta = \arccos(\frac{z}{r}) \end{cases}$$

where f is as in Section B.1.

B.4 Barycentric coordinates

In the n -dimensional affine space $\mathbb{A}^n$, let $P_0, P_1, \dots, P_n$ be points such that $\mathcal{B} = (\overrightarrow{P_0P_1}, \dots, \overrightarrow{P_0P_n})$ is a basis of the direction space $D(\mathbb{A}^n)$. This is the case if and only if the given points are the vertices of an n -simplex. Then $\mathcal{K} = (P_0, \mathcal{B})$ is a Cartesian frame. Every point A has a unique expression of the form

$$\begin{aligned} A &= P_0 + a_1 \overrightarrow{P_0P_1} + \dots + a_n \overrightarrow{P_0P_n} \\ &= (1 - a_1 - \dots - a_n)P_0 + a_1P_1 + \dots + a_nP_n \\ &= \lambda_0P_0 + \lambda_1P_1 + \dots + \lambda_nP_n. \end{aligned}$$

The $n + 1$ values $(\lambda_0, \lambda_1, \dots, \lambda_n)$ are the barycentric coordinates of the point A . Notice that the sum of these coordinates equals 1. The above equation also shows how to translate from Cartesian coordinates to barycentric coordinates. This is the algebraic point of view.

The geometric interpretation in dimension 2 is the following. Let ABC be a triangle of area 1. A point P in the plane of the triangle has barycentric coordinates

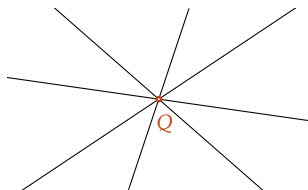
$$\lambda_1 = \text{Area}_{\text{or}}(\triangle PAB), \quad \lambda_2 = \text{Area}_{\text{or}}(\triangle PBC), \quad \lambda_3 = \text{Area}_{\text{or}}(\triangle PCA)$$

where Area_{or} stands for the oriented area of a triangle. Similarly, in higher dimensions, the barycentric coordinates are given by oriented volume relative to an n -simplex.

³Image source: Wikipedia

C.1 Pencil of lines in $\mathbb{A}^2$

Definition C.1. Fix a point $Q \in \mathbb{A}^2$. The set $\mathcal{L}_Q$ of all lines in $\mathbb{A}^2$ passing through Q is called a *pencil of lines*, and Q is called the *center* of the pencil $\mathcal{L}_Q$.



Proposition C.2. If $\ell_1 : a_1x + b_1y + c_1 = 0$ and $\ell_2 : a_2x + b_2y + c_2 = 0$ are two distinct lines in the pencil $\mathcal{L}_Q$, then $\mathcal{L}_Q$ consists of lines having equations of the form

$$\ell_{\lambda,\mu} : \lambda(a_1x + b_1y + c_1) + \mu(a_2x + b_2y + c_2) = 0,$$

where $\lambda, \mu \in \mathbb{R}$ are not both zero. In particular, if $Q = Q(x_0, y_0)$, $\ell_1 : x = x_0$, and $\ell_2 : y = y_0$, then

$$\mathcal{L}_Q = \{ \ell_{\lambda,\mu} : \lambda(x - x_0) + \mu(y - y_0) = 0 : \lambda, \mu \in \mathbb{R} \text{ not both zero} \}.$$

Pencils of lines are useful when a point Q is given as the intersection of two lines, but its coordinates are not known explicitly, and one wants to find the equation of a line passing through Q and satisfying some other conditions. For example, one might require that it contains some point P distinct from Q or that it is parallel to a given line.

Notice that there is redundancy in the two parameters λ and μ , meaning that we don't have two independent parameters here. If $\lambda \neq 0$, then one can divide the equation of $\ell_{\lambda,\mu}$ by λ to obtain

$$\ell_{1,t} : (a_1x + b_1y + c_1) + t(a_2x + b_2y + c_2) = 0,$$

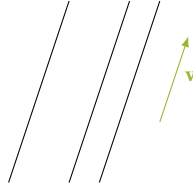
where $t = \frac{\mu}{\lambda} \in \mathbb{R}$. So $\ell_{1,\frac{\mu}{\lambda}}$ and $\ell_{\lambda,\mu}$ are in fact the same line.

Definition C.3. A *reduced pencil* is the pencil $\mathcal{L}_Q$ from which we remove one line, i.e., it is $\mathcal{L}_Q \setminus \{\ell_2\}$ for some $\ell_2 \in \mathcal{L}_Q$. With the above notation and discussion, it is the set

$$\{\ell_{1,t} : (a_1x + b_1y + c_1) + t(a_2x + b_2y + c_2) = 0 : t \in \mathbb{R}\}.$$

The fact that we use one parameter instead of two to describe almost all lines passing through Q simplifies calculations.

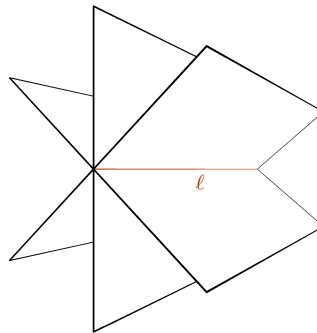
Definition C.4. Let $\mathbf{v} \in \mathbb{V}^2$. The set $\mathcal{L}_{\mathbf{v}}$ of all lines in $\mathbb{A}^2$ with direction vector $\mathbf{v}$ is called an *improper pencil of lines*, and $\mathbf{v}$ is called a *direction vector* of the pencil $\mathcal{L}_{\mathbf{v}}$.



The connection between pencils of lines and improper pencils of lines is best understood through projective geometry, where the improper pencil of lines is the set of all lines intersecting at the same point at infinity.

C.2 Pencil of planes in $\mathbb{A}^3$

Definition C.5. Let $\ell \subseteq \mathbb{A}^3$ be a line. The set Π_{ℓ} of all planes in $\mathbb{A}^3$ containing ℓ is called a *pencil of planes*, and ℓ is called the *axis* (or *carrier line*) of the pencil Π_{ℓ} .



Proposition C.6. If $\pi_1 : a_1x + b_1y + c_1z + d_1 = 0$ and $\pi_2 : a_2x + b_2y + c_2z + d_2 = 0$ are two distinct planes in the pencil Π_ℓ , then Π_ℓ consists of planes having equations of the form

$$\pi_{\lambda,\mu} : \lambda(a_1x + b_1y + c_1z + d_1) + \mu(a_2x + b_2y + c_2z + d_2) = 0,$$

where $\lambda, \mu \in \mathbb{R}$ are not both zero.

Pencils of planes are useful when a line ℓ is given as the intersection of two planes (see Subsection 2.3.2) and one wants to find the equation of a plane containing ℓ and satisfying some other conditions. For example, one might require that it contains some point P which does not belong to ℓ or that it is parallel to a given line.

As in the case of line pencils, there is redundancy in the two parameters λ and μ . If $\lambda \neq 0$, then one can divide the equation of $\pi_{\lambda,\mu}$ by λ to obtain

$$\pi_{1,t} : (a_1x + b_1y + c_1z + d_1) + t(a_2x + b_2y + c_2z + d_2) = 0,$$

where $t = \frac{\mu}{\lambda} \in \mathbb{R}$. So $\pi_{1,\frac{\mu}{\lambda}}$ and $\pi_{\lambda,\mu}$ are in fact the same plane.

Definition C.7. A *reduced pencil* is the pencil Π_ℓ from which we remove one plane, i.e., it is $\Pi_\ell \setminus \{\pi_2\}$ for some $\pi_2 \in \Pi_\ell$. With the above notation and discussion, it is the set

$$\left\{ \pi_{1,t} : (a_1x + b_1y + c_1z + d_1) + t(a_2x + b_2y + c_2z + d_2) = 0 : t \in \mathbb{R} \right\}.$$

The fact that we use one parameter instead of two to describe almost all planes containing ℓ simplifies calculations.

Definition C.8. Let $\mathbb{W} \subseteq \mathbb{V}^3$ be a vector subspace of dimension 2. The set $\Pi_{\mathbb{W}}$ of all planes in $\mathbb{A}^3$ which admit $\mathbb{W}$ as direction space is called an *improper pencil of planes*, and $\mathbb{W}$ is called the vector subspace associated with the pencil $\Pi_{\mathbb{W}}$.

The connection between pencils of planes and improper pencils of planes is best understood through projective geometry, where we can think of the improper pencil of planes as the set of all planes intersecting along a line at infinity.

Some classical theorems in affine geometry

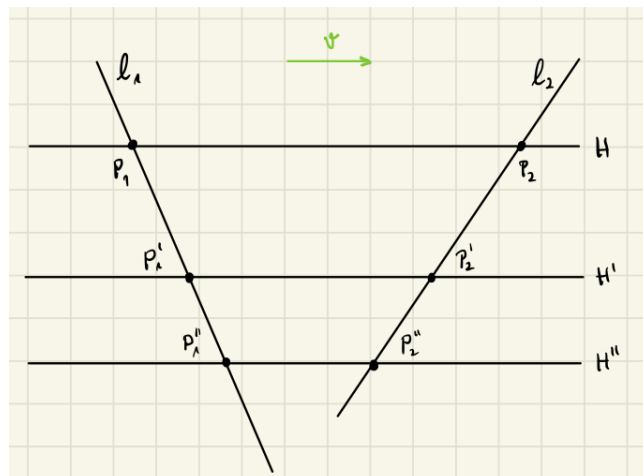
Theorem D.1 (Thales' intercept theorem). Let $H, H',$ and H'' be three distinct parallel lines in the affine plane $\mathbb{A}^2$. Let ℓ_1 and ℓ_2 be two lines not parallel to $H, H',$ and H'' . For $i = 1, 2$, let

$$\begin{aligned} P_i &= \ell_i \cap H \\ P'_i &= \ell_i \cap H' \\ P''_i &= \ell_i \cap H'' \end{aligned}$$

and let k_1, k_2 be scalars such that

$$\overrightarrow{P_i P''_i} = k_i \overrightarrow{P_i P'_i}.$$

Then, $k_1 = k_2$.

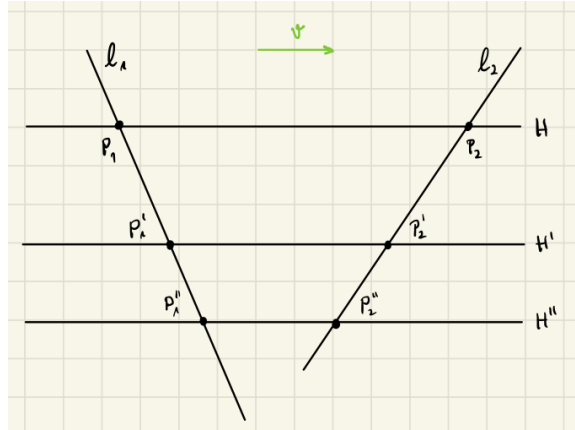


Theorem D.2 (Converse of Thales' intercept theorem). Let ℓ_1 and ℓ_2 be two lines in the affine plane $\mathbb{A}^2$. For $i = 1, 2$, let P_i, P'_i , and P''_i be three distinct points on ℓ_i . Let H be the line passing through P_1 and P_2 , and let H' be the line passing through P'_1 and P'_2 .

Suppose that H is parallel to H' , and let k be a scalar such that for $i = 1, 2$,

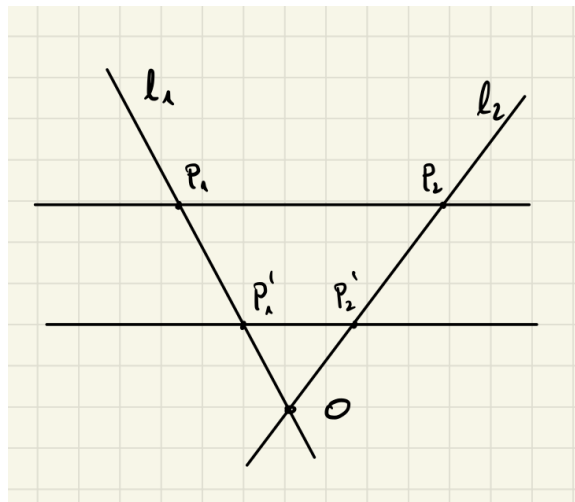
$$\overrightarrow{P_i P''_i} = k \overrightarrow{P_i P'_i}.$$

Then, the line H'' passing through P''_1 and P''_2 is parallel to H and H' .

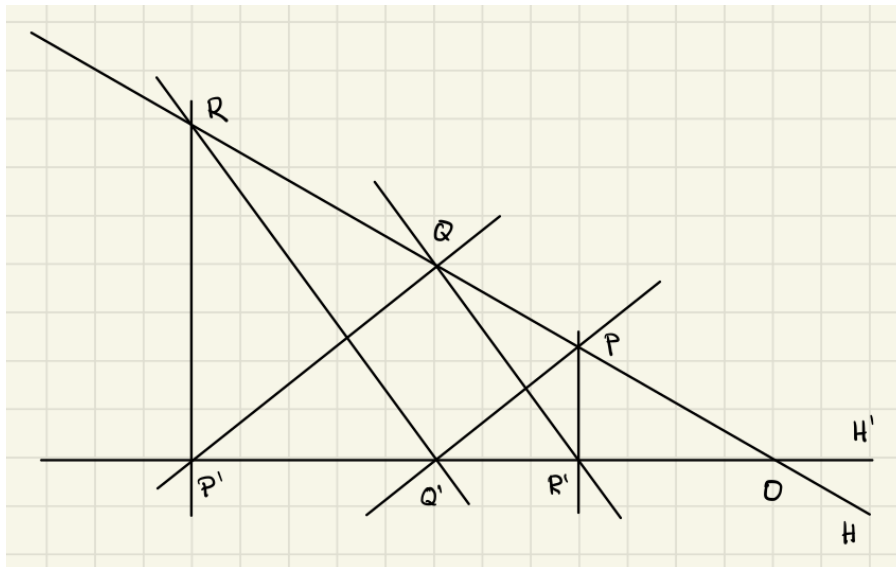


Theorem D.3 (Thales' theorem for triangles). Let ℓ_1 and ℓ_2 be two lines in the affine plane $\mathbb{A}^2$ intersecting at a point O . For $i = 1, 2$, let P_i and P'_i be points on ℓ_i distinct from O . Then the line $P_1 P_2$ is parallel to the line $P'_1 P'_2$ if and only if there exists a scalar k such that

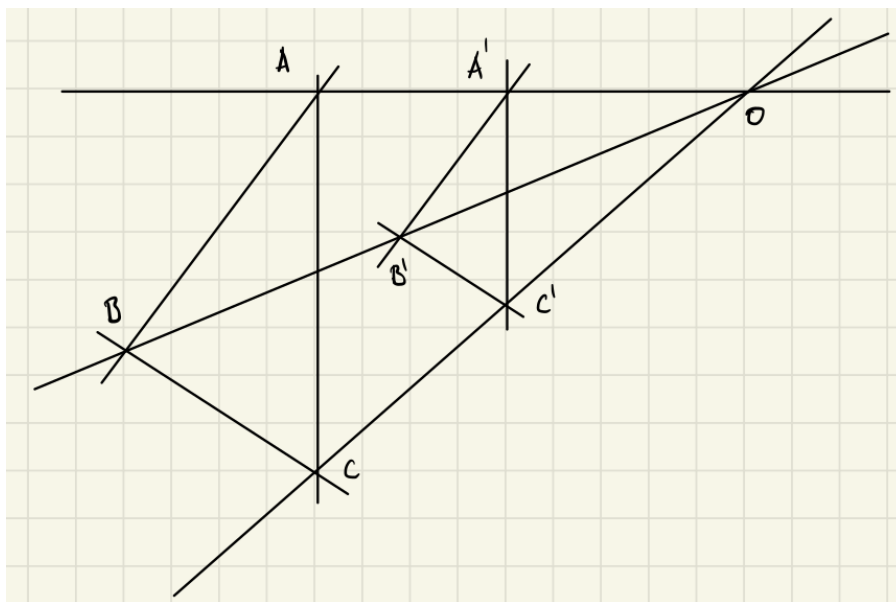
$$\overrightarrow{OP'_i} = k \overrightarrow{OP_i} \quad \text{for } i = 1, 2.$$



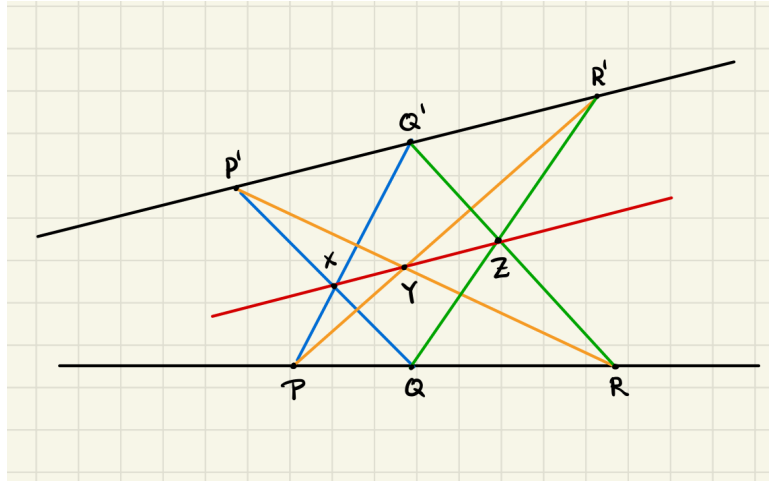
Theorem D.4 (Pappus' affine theorem). Let H and H' be two distinct lines in the affine plane $\mathbb{A}^2$. Let $P, Q, R \in H$ and $P', Q', R' \in H'$ be distinct points, none of which lies at the intersection $H \cap H'$. If $PQ' \parallel P'Q$ and $QR' \parallel Q'R$, then $PR' \parallel P'R$.



Theorem D.5 (Desargues' theorem). Let $A, B, C, A', B', C' \in \mathbb{A}^2$ be points such that no three are collinear, and such that $AB \parallel A'B'$, $BC \parallel B'C'$, and $AC \parallel A'C'$. Then the three lines AA' , BB' , and CC' are either parallel or have a point in common.



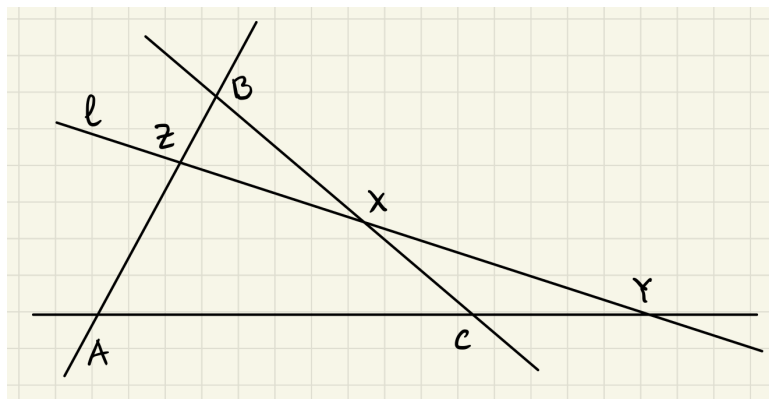
Theorem D.6 (Pappus' hexagon theorem). Let H and H' be two distinct lines in the affine plane $\mathbb{A}^2$. Let $P, Q, R \in H$ and $P', Q', R' \in H'$ be distinct points, none of which lies at the intersection $H \cap H'$. Assume that $PQ' \cap P'Q = \{X\}$, $PR' \cap P'R = \{Y\}$, and $QR' \cap Q'R = \{Z\}$. Then the points X, Y, Z are collinear.



Theorem D.7 (Newton-Gauss line). A complete quadrilateral is the configuration of four lines, no three of which pass through the same point. Let O, A, B, C, B', A' be the intersection points of such lines, i.e., the vertices of the complete quadrilateral, such that A lies between O and B and such that A' lies between O and B' . The diagonals of this quadrilateral are the segments $[OC]$, $[AA']$, and $[BB']$. The midpoints of the diagonals of a complete quadrilateral are collinear.

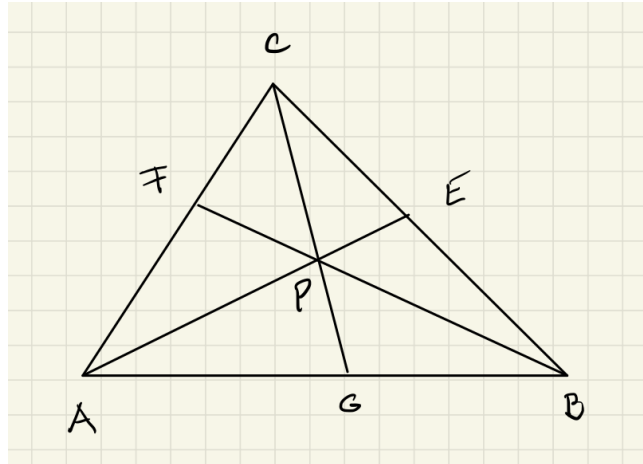
Theorem D.8 (Menelaus' theorem). Let ABC be a triangle and ℓ a line which does not pass through the vertices of the triangle. Let X, Y, Z be points on the lines BC, CA , and AB , respectively, and consider the signed ratios $\alpha = BX : XC$, $\beta = CY : YA$, and $\gamma = AZ : ZB$. Then X, Y, Z are collinear if and only if

$$\alpha \cdot \beta \cdot \gamma = -1.$$

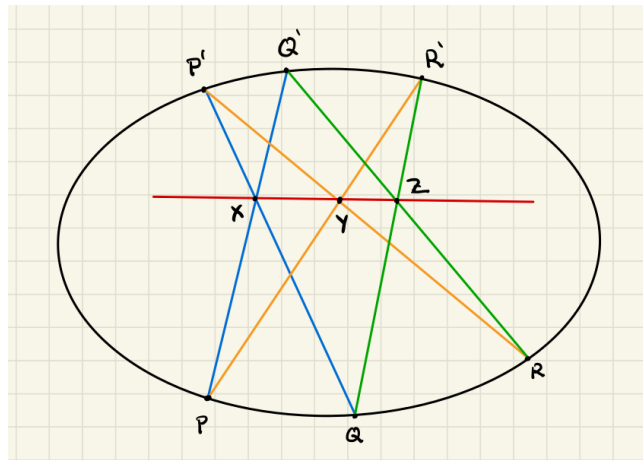


Theorem D.9 (Ceva's theorem). Let ABC be a triangle. Let $E \in BC$, $F \in CA$, and $G \in AB$ and consider the signed ratios $\alpha = BE : EC$, $\beta = CF : FA$, and $\gamma = AG : GB$. The three lines AE , BF , and CG are concurrent if and only if

$$\alpha \cdot \beta \cdot \gamma = 1.$$



Theorem D.10 (Pascal's hexagrammum mysticum theorem). If all six vertices of a hexagon lie on a conic section and the three pairs of opposite sides intersect, then the three points of intersection are collinear.



- [1] C. Alsina, B. Nelsen, A mathematical space odyssey. Solid geometry in the 21st century. The Dolciani Mathematical Expositions, 50. Mathematical Association of America, Washington, DC, 2015.
- [2] J. Benítez, A unified proof of Ceva and Menelaus' theorems using projective geometry. J. Geom. Graph. 11 (2007), no. 1, 39–44.
- [3] M. de Berg, O. Cheong, M. van Kreveld, M. Overmars, Computational geometry. Algorithms and applications. Third edition. Springer-Verlag, Berlin, 2008.
- [4] M. Berger, Geometry I. Translated from the 1977 French original by M. Cole and S. Levy. Fourth printing of the 1987 English translation Universitext. Springer-Verlag, Berlin, 2009.
- [5] P.A. Blaga, Geometrie liniară. Cu un ochi către grafica pe calculator, Vol. I, Presa Universitară Clujeană, 2022.
- [6] C.B. Boyer, A history of mathematics. Second edition. Edited and with a preface by Uta C. Merzbach. John Wiley & Sons, Inc., New York, 1989.
- [7] H.S.M. Coxeter, Regular polytopes. Methuen, London, 1948.
- [8] H.S.M. Coxeter, S.L. Greitzer, Geometry revisited. New Mathematical Library, 19. Random House, Inc., New York, 1967.
- [9] S. Crivei, Basic Linear Algebra. Presa Universitară Clujeană, 2022.
- [10] Euclid. The thirteen books of Euclid's Elements translated from the text of Heiberg. Vol. I: Introduction and Books I, II. Vol. II: Books III–IX. Vol. III: Books X–XIII and Appendix. Translated with introduction and commentary by Thomas L. Heath. 2nd ed. Dover Publications, Inc., New York, 1956.
- [11] A.E. Fekete, Real linear algebra. Monographs and Textbooks in Pure and Applied Mathematics, 91. Marcel Dekker, Inc., New York, 1985.

- [12] I. Grattan-Guinness, The search for mathematical roots, 1870–1940. Logics, set theories and the foundations of mathematics from Cantor through Russell to Gödel. Princeton Paperbacks. Princeton University Press, Princeton, NJ, 2000.
- [13] J. Hefferon, Linear Algebra, 2020.
- [14] D. Hilbert, Foundations of Geometry. Second English edition. Translated by Unger L. from 10th ed. Revised and Enlarged by Bernays P. Open Court, Illinois, 1971.
- [15] L. Hodgkin, A history of mathematics. From Mesopotamia to modernity. Oxford University Press, Oxford, 2005.
- [16] J.M. Lee, Axiomatic Geometry. American Mathematical Society, 2013.
- [17] M. Nechita, Lecture notes for mathematical analysis, 2025.
- [18] C.C. Pugh, Real mathematical analysis. Second edition. Undergraduate Texts in Mathematics. Springer, Cham, 2015.
- [19] W. Rudin, Principles of mathematical analysis. Third edition. International Series in Pure and Applied Mathematics. McGraw-Hill Book Co., New York-Auckland-Düsseldorf, 1976.
- [20] E. Sernesi, Linear Algebra. A geometric approach. Translated by Montaldi J., Chapman & Hall/CRC, 1993.
- [21] Handbook of discrete and computational geometry. Second edition. Edited by Jacob E. Goodman and Joseph O'Rourke. Discrete Mathematics and its Applications (Boca Raton). Chapman & Hall/CRC, Boca Raton, FL, 2004.
- [22] Lean 4, Programming Language and Theorem Prover, <https://lean-lang.org/>
- [23] The mathlib Community: The Lean mathematical library. In: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs. p. 367–381. CPP 2020, New York, NY, USA (2020).
- [24] A mathlib overview, <https://leanprover-community.github.io/mathlib-overview.html>